

Compatible Role-Occupancy Closure

for the Neutrino Propagation Role Complex

Universal compatible-occupancy lifting, carrier-semantic descent, and terminal over-residue

Main theorem

Let $\mathfrak{S}_\nu^{(4)}$ be the exact PCR-4 carrier-survivor quotient. Every survivor has at least one complete internally eligible and compatible maximal-factorization assignment, so the factorization-sensitive internal orbit object has nonempty carrier fibers:

$$\mathfrak{D}_5^{\text{int}} = \bigsqcup_{[x] \in \mathfrak{S}_\nu^{(4)}} \mathfrak{F}_5([x]), \quad \mathfrak{F}_5([x]) \neq \emptyset.$$

At terminal carrier resolution, factorization alternatives over one parent are quotiented by the carrier projection:

$$\mathfrak{S}_{\nu, \text{car}}^{(5)} := \text{im}(\bar{\pi}_{\text{car}}) = \mathfrak{S}_\nu^{(4)}, \quad |\mathfrak{S}_{\nu, \text{car}}^{(5)}| \geq 2.$$

A law-only Weinberg carrier and an active three-singlet carrier remain distinct after every applicable same-target import in the bounded reference corpus, hence

$$|\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2.$$

The terminal carrier-resolution target has one complete-carrier role $R_5^{\text{term}} = \{r_{\text{car}}(T_\nu^{\text{osc}})\}$. Its eligibility relation E_5^{term} , compatibility predicate J_5^{term} , and preassignment union give

$$\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] = \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+], \quad |\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+]| \geq 2 > 1 = |R_5^{\text{term}}|.$$

Therefore no compatible bijective role-covering assignment exists and, under the complete terminal ROC certificate,

$$\neg \text{RealizationLicensed}(\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)}).$$

The terminal diagnosis is over-residue after the strongest factorization-insensitive carrier quotient supported by the fixed theorem chain.

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Abstract

PCR–1 through PCR–4 fix an ordered neutrino propagation role complex, a complete target-facing interface, a source-first carrier universe, and the exact carrier-semantic survivor residue under simultaneous role, channel, response, gauge, anomaly, vacuum, effective-theory, extension, and common-cone constraints. PCR–5 asks whether compatible role occupancy reduces that residue to a unique physical carrier realization.

The analysis separates internal occupancy structure from carrier identity. For each PCR–4 survivor x , finite witness-preserving normalization produces at least one normal form whose mandatory witness paths define a nonempty role-bearing core. The core has a finite nonempty poset of admissible factorizations; every factor block has nonempty role-standing signature, all maximal factorizations are retained, and no canonical common refinement is assumed. Every maximal factorization admits a full-factor assignment passing pointwise E_5 and global J_5 . Hence every PCR–4 carrier class has a nonempty internal occupancy fiber and

$$|\mathfrak{D}_5^{\text{int}}| \geq |\mathfrak{S}_\nu^{(4)}| \geq 2.$$

Thus compatible occupancy universally lifts the PCR–4 carrier residue.

The factorization-sensitive internal quotient is then projected to carrier-semantic resolution. Alternative maximal factorizations over one carrier collapse under $\bar{\pi}_{\text{car}}$, giving

$$\mathfrak{S}_{\nu, \text{car}}^{(5)} = \text{im}(\bar{\pi}_{\text{car}}) = \mathfrak{S}_\nu^{(4)}.$$

A law-only Weinberg carrier and a constructive full-rank active SM + $3\nu_R$ carrier remain distinct because their parent carrier traces differ. The bounded cross-corpus analysis leaves both carrier classes live, so $|\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2$.

At terminal carrier resolution, the fixed one-role target is evaluated using a separately formed eligibility relation E_5^{term} and compatibility predicate J_5^{term} . The union of all positive terminal preassignments is $\mathfrak{S}_\nu^{(4)}$ natively and $\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$ under the bounded update. Since the live carrier residue contains at least two classes while the terminal role set has one element, no compatible bijective role-covering assignment exists. The role-realization-complete ROC certificate therefore yields

$$\neg \text{RealizationLicensed}(\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)}).$$

The endpoint is a lower bound, not an exact two-carrier theorem; it neither assumes a unique maximal factorization nor excludes later target extensions carrying genuinely new ultraviolet-sensitive information.

Keywords. admissibility; neutrino oscillation; role occupancy; maximal factorization; universal lifting; carrier-semantic residue; terminal eligibility; terminal compatibility; over-residue; effective field theory; right-handed neutrino; ultraviolet origin; AASC.

Scope and interpretation

The theorem is exact relative to the fixed PCR–1–PCR–4 imports, the finite PCR–3 source scope, the independently generated PCR–5 internal grammar, the declared terminal carrier-resolution target, and the bounded reference corpus. The internal factorization-sensitive quotient is not identified with the terminal physical carrier quotient; terminal multiplicity is established only after factorization alternatives over one carrier are collapsed. Supplementary provenance and consistency records support reproducibility but do not substitute for the semantic normalization, lifting, carrier-descent, terminal handoff, or ROC proofs.

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1 Introduction

1.1 The remaining problem after PCR-4

PCR-4 leaves a proper nonempty carrier-semantic survivor quotient $\mathfrak{S}_v^{(4)}$ containing at least two carrier-inequivalent classes. That result proves simultaneous survival, not compatible occupancy closure. A survivor may realize the ordered role packet through one irreducible region, distinct nested regions, or a distributed finite diagram, and one carrier point may admit several incomparable maximal factorizations. PCR-5 therefore asks two distinct questions: whether every carrier survivor has at least one complete compatible occupancy architecture, and whether the resulting *carrier-semantic* residue closes to one terminal realization.

1.2 Structural distinctions required by role occupancy

Four distinctions are essential to the analysis.

1. An internal assignment must cover its complete role-bearing factorization rather than a selected subset.
2. A PCR-4 common cone proves coherent co-instantiation but does not determine the number of occupation blocks.
3. Alternative admissible factorizations are retained unless a lawful equivalence identifies them; no canonical common refinement is assumed.
4. Factorization-sensitive internal orbits reach terminal ROC only after carrier-semantic descent and the independent formation of terminal eligibility and compatibility.

These distinctions keep internal decomposition, physical carrier identity, and terminal role occupation at their appropriate resolutions.

1.3 Three levels of the analysis

The calculation has three typed levels.

1. *Internal occupancy*: all complete maximal-factorization assignments are formed inside each PCR-4 survivor.
2. *Carrier-semantic descent*: factorization-sensitive internal orbits are projected to the fixed PCR-4 carrier quotient, so alternative decompositions of one carrier do not generate spurious physical-carrier multiplicity.
3. *Terminal ROC*: on the fixed one-role carrier-resolution target, $E_5^{\text{term}}, J_5^{\text{term}}$, the compatible preassignment space, and the live residue are formed before testing for a compatible bijection.

Every target and role type is fixed before the downstream residue cardinalities are evaluated.

1.4 Role-bearing cores and maximal factorizations

For each irreducible normal form N , $\text{Core}_5(N)$ is the source-constitutive closure of mandatory PCR-5 role/interface witness paths. Auxiliary parent-carrier, continuation, parameter, and failure data remain in $\text{Aux}_5(N)$ unless they themselves lie on a mandatory role-standing path. An admissible factorization covers the whole role core, reconstructs the witness diagram, retains all constitutive maps, and requires every block to have nonempty role-standing signature. The one-block factorization proves nonemptiness; finite-poset maximality supplies at least one maximal factorization. Several incomparable maximal factorizations may coexist and are all retained internally.

1.5 Universal compatible-occupancy lifting

For every survivor x , choose any $N \in \mathcal{N}_5(x)$ and any $F \in \text{MaxFact}_5(N)$. Every factor block passes E_5 by admissibility, and the complete diagram passes J_5 by preserved PCR-4 common-cone coherence. Hence

$$\Sigma_5^{\text{int}}(x) \neq \emptyset \quad \forall x \in \mathfrak{S}_v^{(4)}. \quad (1)$$

The internal factorization-sensitive object therefore decomposes into nonempty carrier fibers:

$$\mathfrak{D}_5^{\text{int}} = \bigsqcup_{[x] \in \mathfrak{S}_\nu^{(4)}} \mathfrak{F}_5([x]), \quad |\mathfrak{D}_5^{\text{int}}| \geq |\mathfrak{S}_\nu^{(4)}| \geq 2. \quad (2)$$

Compatible occupancy therefore preserves every PCR-4 carrier class; the internal result is a universal lifting theorem.

1.6 Carrier-semantic quotient and explicit separation

The terminal carrier target does not distinguish maximal factorizations over one parent merely because their internal graphs differ. Define

$$\mathfrak{S}_{\nu, \text{car}}^{(5)} := \text{im}(\bar{\pi}_{\text{car}}). \quad (3)$$

Universal lifting gives $\mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_\nu^{(4)}$. A law-only Weinberg survivor x_W and a generic active three-singlet survivor x_{3N} have complete internal assignments but unequal fixed carrier traces,

$$\chi_{\text{car}}(x_W) \not\equiv \chi_{\text{car}}(x_{3N}),$$

so their carrier-semantic residues remain distinct after factorization distinctions within each carrier are forgotten.

1.7 Cross-corpus stability

The bounded reference corpus is applied only through predicates well typed on the unchanged PCR-5 target. Fixed-load Standard Model closure is not propagated into a different target; generation number is not identified with singlet number; low-energy exactness is not promoted to ultraviolet carrier identity; and EFT matching is not treated as a field-inventory isomorphism. If

$$\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+] := \bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]),$$

then the two benchmark carrier classes survive and

$$\boxed{|\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2.} \quad (4)$$

1.8 Terminal ROC theorem

The terminal carrier-resolution target has the fixed role architecture

$$R_5^{\text{term}} = \{r_{\text{car}}(T_\nu^{\text{osc}})\}.$$

This singleton role is a field of the target architecture; it is neither inferred from the singularity of the underlying target instance nor chosen to match the residue count. PCR-5 defines a terminal eligibility relation E_5^{term} from same-target carrier provenance and a terminal compatibility predicate J_5^{term} from the existence of a complete internally compatible occupancy witness. Therefore

$$\text{Residue}_{\text{term}}^{\text{live}} = \mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_\nu^{(4)}, \quad \text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] = \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+].$$

Since both live residues have at least two carrier classes while $|R_5^{\text{term}}| = 1$, no compatible bijective role-covering assignment exists. The complete terminal ROC certificate gives

$$\boxed{\neg \text{RealizationLicensed}(\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)}).} \quad (5)$$

1.9 Strength and limitation

PCR-5 does not prove exactly two carrier classes, unique maximal factorization within a carrier fiber, or impossibility of a later UV-sensitive selector. It proves that every PCR-4 carrier survivor admits compatible occupancy and that, even after factorization-sensitive multiplicity is collapsed to the strongest fixed carrier-semantic quotient and the bounded same-target corpus is applied, at least two carrier classes remain live.

1.10 Organization

Sections 2–4 specify the fixed dependencies and target types. Sections 5–14 develop role-bearing factorizations and prove universal complete-assignment existence. Sections 15–16 distinguish the factorization-sensitive internal quotient from carrier-resolution descent. Sections 17–22 establish carrier-indexed lifting and the internal lower bound. Section 23 proves bounded carrier-semantic stability. Section 24 constructs terminal E/J , the live residue, certificate completeness, and the ROC theorem.

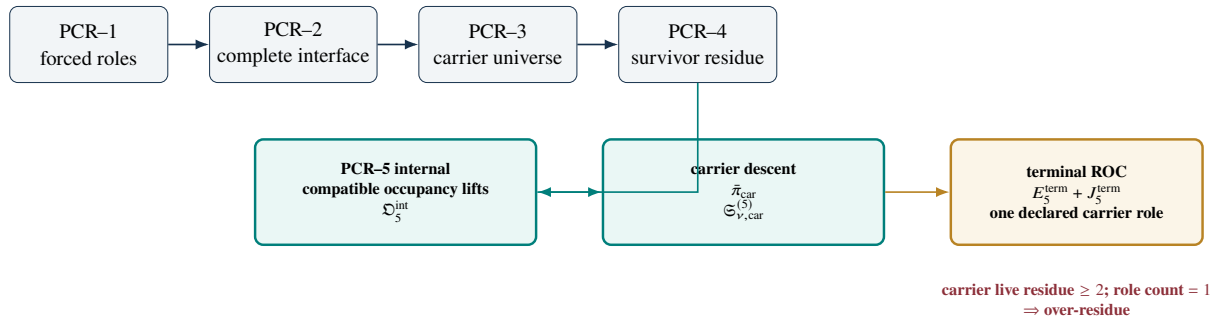


Figure 1: PCR-5 constructs compatible internal occupancy, descends to carrier-semantic residue, and then forms terminal ROC eligibility, compatibility, and live residue.

2 Fixed dependencies and scope discipline

Table 1: Fixed PCR-5 dependencies.

Source	Imported object	Permitted use in PCR-5
PCR-1	$\mathcal{R}_\nu^{\text{osc}} = (\{r_{\triangleright\Delta}, r_\Delta\}, r_\Delta \succ r_{\triangleright\Delta})$	Fixes role definitions, witnesses, and strict order; does not fix occupant number or carrier identity.
PCR-2	$\mathfrak{S}_\nu^{\text{adm}}, \mathcal{P}_\nu^+, \mathcal{J}_\nu^{\text{sim}}$	Fixes the complete target-facing interface and lawful interface redescription; does not identify microscopic carriers.
PCR-3	$\mathfrak{U}_\nu, \chi_{\text{car}}, \equiv_{\text{car}}$	Fixes source-first carrier generation, pointwise activity reduction, spectator stripping, and carrier identity.
PCR-4	$\mathfrak{S}_\nu^{(4)}, \mathcal{W}_4(x)$	Fixes the exact simultaneous survivor residue and one complete one-source witness/common-cone packet for each survivor.
ROC	Forced role set, role eligibility, joint compatibility, preassignment, live-residue, role-covering bijection, and realization-licensing theorem	Applied only at the separately declared terminal carrier-resolution target; internal E_5/J_5 are not substituted for terminal $E_5^{\text{term}}/J_5^{\text{term}}$.

The fixed dependency packet is

$$\mathfrak{F}_5 = (T_{\nu}^{\text{osc}}, \mathcal{R}_{\nu}^{\text{osc}}, \mathfrak{S}_{\nu}^{\text{adm}}, \mathfrak{U}_{\nu}, \mathfrak{S}_{\nu}^{(4)}, \equiv_{\text{car}}, \mathbb{S}_4^{\text{surv}}). \quad (6)$$

Adding a heavy-neutral-lepton record, neutrinoless double-beta-decay standing, correlated nonunitarity, a cosmological mass sum, a fully activated global $B - L$ target, or another UV-sensitive coordinate changes the target and therefore lies outside the fixed PCR-5 problem.

Definition 2.1 (PCR-5 import lock). *A construction is import faithful when it preserves the PCR-1 role complex, PCR-2 target fiber, PCR-3 carrier identity, PCR-4 survivor/common-cone packets, and the ROC requirement that the terminal role set be fixed before the live residue is formed.*

Theorem 2.2 (No assignment backflow). *Neither the cardinality nor interpretation of $\mathfrak{S}_5^{\text{int}}$ or $\text{Residue}_{\text{term}}^{\text{live}}$ may alter the frozen packet, normalization grammar, role-core rule, factorization predicate, evaluators, carrier invariant, or terminal role set.*

Proof. Those objects define what is normalized, factored, assigned, compared, and licensed. Allowing the endpoint to rewrite them would let a candidate participate in its own admissibility or equivalence and would reverse the dependency order. $\square$

2.1 Logical polarity

PCR-1 is role necessity; PCR-2 is interface completion; PCR-3 is carrier generation and identity; PCR-4 is simultaneous survival; PCR-5 is internal role-bearing factorization plus terminal closure. A PCR-4 common cone proves co-instantiation and global compatibility. It proves neither one cell nor one maximal factorization. A complete internal orbit is not automatically a terminal residue. It must first descend to carrier semantics, receive a terminal eligibility edge, and pass the separately defined terminal compatibility predicate.

Theorem 2.3 (Dependency sufficiency for the final calculation). *The frozen packet, finite witness-preserving normalization, role-bearing-core extraction, finite maximal-factorization construction, factorization-relative evaluators, carrier projection, separately declared terminal role architecture, and terminal eligibility/compatibility relations contain all data required to decide the internal lifting result and the terminal balanced/under-residue/over-residue branch at the declared scope.*

Proof. The upstream stages fix roles, interface, carrier domain, identity, survivors, and complete witness diagrams. PCR-5 supplies the missing role-bearing decomposition and complete internal assignment data. The carrier projection then removes factorization overcount at terminal resolution. Finally the terminal stage constructs the ROC eligibility relation, compatibility predicate, preassignment set, and live residue before testing the fixed one-role carrier target. $\square$

No dependency laundering

Standardness, economy, ultraviolet motivation, or familiarity does not promote a carrier. Conversely, a future discriminator cannot be inserted after the multiplicity result and described as part of the original target.

3 Two-level target and certificate architecture

3.1 Internal occupancy target

Define

$$T_{\nu, \text{int}}^{(5)} = (T_{\nu}^{\text{osc}}, \mathcal{R}_{\nu}^{\text{osc}}, \mathfrak{S}_{\nu}^{\text{adm}}, \mathfrak{S}_{\nu}^{(4)}, \text{OneSurvivor}, \text{FiniteSupport}, \mathcal{G}_5, \text{Core}_5, \text{Fact}_5, \text{LawfulRedescription}). \quad (7)$$

This target asks for complete role-bearing factorization architectures inside one PCR-4 survivor. It does not yet ask which carrier-semantic residue occupies the terminal carrier role.

Definition 3.1 (Internal comparison class). *An object lies in $\mathfrak{C}_{\nu, \text{int}}^{(5)}$ when it uses one $x \in \mathfrak{S}_{\nu}^{(4)}$, forms its support from the finite PCR-4 witness diagram and source-constitutive closure, normalizes only by rules in $\mathcal{G}_5$, extracts $\text{Core}_5(N)$ from mandatory witness paths, factors only by the admissibility clauses of $\text{Fact}_5(N)$, preserves $\chi_{\text{car}}(x)$, and evaluates all role, channel, response, boundary, continuation, parameter, and failure data at one source point.*

3.2 Terminal carrier-resolution target

The terminal target is fixed *before* any internal orbit or residue cardinality is inspected. It is the one-role carrier-resolution question

Which carrier-semantic realization class occupies the complete-carrier slot of the frozen neutrino propagation target?

Accordingly,

$$\begin{aligned} T_{\nu, \text{term}}^{(5)} &= (T_{\nu}^{\text{osc}}, \text{CarrierResolution}, \text{OneCompleteCarrierRole}, \\ &\quad \text{CarrierFaithfulResidue}), \\ R_5^{\text{term}} &= \{r_{\text{car}}(T_{\nu}^{\text{osc}})\}. \end{aligned} \tag{8}$$

The singleton role is therefore part of the declared terminal target architecture; it is not inferred from the number of target instances and is not chosen to match the later residue count.

Theorem 3.2 (Terminal role-set completeness). *For the separately declared target $T_{\nu, \text{term}}^{(5)}$,*

$$R_{T_{\nu, \text{term}}^{(5)}}(\mathfrak{C}_{\nu, \text{term}}^{(5)}) = R_5^{\text{term}} = \{r_{\text{car}}(T_{\nu}^{\text{osc}})\}, \quad |R_5^{\text{term}}| = 1,$$

and this role set is fixed before $\mathfrak{D}_5^{\text{int}}$, $\mathfrak{S}_{\nu, \text{car}}^{(5)}$, or $\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$ is calculated.

Proof. The role architecture of $T_{\nu, \text{term}}^{(5)}$ contains exactly one declared use: occupation of the complete-carrier slot at carrier-semantic resolution. The Forced Role Set theorem of Role-Occupancy Closure therefore returns this one role. The conclusion uses target architecture rather than target-instance cardinality. Because (8) is fixed upstream of the internal count, no cardinality backfitting occurs. $\square$

Proposition 3.3 (Role-set independence). *The same singleton terminal role set is present whether the later carrier-semantic live residue is empty, singleton, or multiple.*

Proof. The role set is a field of the already declared terminal target. Residue cardinality is a downstream calculation and cannot alter that field without replacing the target. $\square$

3.3 Certificate structure

The certificate packet is

$$\mathfrak{C}_{\nu, \text{cert}}^{(5)} = \left(\begin{array}{l} T_{\nu, \text{int}}^{(5)}, T_{\nu, \text{term}}^{(5)}, R_5, R_5^{\text{term}}, \mathcal{G}_5, \mathcal{N}_5, \\ \text{Core}_5, \text{Aux}_5, \text{Fact}_5, \text{MaxFact}_5, \Omega_5, \mathfrak{P}_5^{\text{adm}}, \\ E_5, J_5, \Sigma_5^{\text{int}}, \Gamma_5^{\text{int}}, \mathfrak{D}_5^{\text{int}}, \mathfrak{S}_{\nu, \text{car}}^{(5)}, \\ E_5^{\text{term}}, J_5^{\text{term}}, \text{Residue}_{\text{term}}^{\text{elig}}, \Sigma_5^{\text{pre, term}}, \text{Residue}_{\text{term}}^{\text{live}}, \Sigma_5^{\text{term}}, I_5^{\text{term}}, \Gamma_5^{\text{term}}, \\ \text{Complete}_{\mathcal{G}_5}, \text{Complete}_{\text{term}}, \text{CorpusBound}_{\mathcal{K}_+} \end{array} \right). \tag{9}$$

Definition 3.4 (Grammar-relative internal completeness). *Complete $_{\mathcal{G}_5}$ holds when:*

1. $\mathcal{G}_5$ is generated from the frozen target and upstream witness sorts before survivor inspection;
2. every finite raw witness support terminates in at least one witness-preserving normal form;
3. every normal form has a nonempty role-bearing core and auxiliary trace complement;
4. every role-bearing core has at least one maximal admissible factorization;
5. every factor block has nonempty role-standing signature;
6. every assignment covers the entire selected maximal factorization;
7. E_5 and J_5 are total on the objects generated by $\mathcal{G}_5$;
8. Γ_5^{int} removes only $\mathcal{G}_5$ -lawful factorization-sensitive presentation differences;
9. every PCR-4 survivor has a nonempty internal orbit fiber.

This is not an unrestricted claim about all future physical distinctions, nor does it assert that two incomparable maximal factorizations over one parent carrier are distinct carrier-semantic realizations.

3.4 Complete internal trace

For

$$z = (x, N, F, P_{N,F}, \sigma, e, j) \in \Sigma_5^{\text{int}},$$

set

$$\tau_{\text{asg}}^{(5)}(z) = \left(\begin{array}{c} T_{v,\text{int}}^{(5)}, \mathcal{G}_5, \chi_{\text{car}}(x), N, \text{Core}_5(N), \text{Aux}_5(N), F, \Omega_5(N, F), \\ G_{N,F}, I_{N,F}, P_{N,F}, \sigma, \{\tau_{\text{occ}}^{(5)}(\omega; x)\}_{\omega \in \Omega_5(N,F)}, e, j, \\ \mathcal{B}_{\text{CC}}, \mathcal{P}_v^+, \mathcal{J}_v^{\text{sim}}, \text{parameter stratum, boundary, continuation, and matching data,} \\ \text{extension and failure data} \end{array} \right). \quad (10)$$

Auxiliary data are retained in the complete invariant even when they are not role-bearing factor blocks.

Proposition 3.5 (Interface equality is not realization identity). *Equal PCR-2 target-interface data do not imply equal parent carrier-semantic residues.*

Proof. PCR-3 and PCR-4 retain carrier-distinct survivors with the same target-facing role packet. Primitive-node inventory, source incidence, continuation, and failure domains remain fields of the frozen carrier trace and of (10). $\square$

3.5 Completeness at distinct resolutions

The manuscript distinguishes: (i) upstream semantic candidate completeness, proved in PCR-3 and PCR-4; (ii) internal factorization-sensitive completeness relative to $\mathcal{G}_5$, proved here; and (iii) bounded corpus disposition relative to $\mathcal{K}_+$. The terminal carrier-resolution certificate is constructed separately in section 24. No finite registry, factorization ledger, or trace definition is promoted into unrestricted future semantic exhaustion.

4 Internal role structure and the terminal carrier role

4.1 Inherited physical role complex

The internal physical work remains

$$R_5 = \{r_{\triangleright\Delta}, r_{\Delta}\}, \quad r_{\Delta} \succ_R r_{\triangleright\Delta}. \quad (11)$$

The weaker role records charged-current boundary nonpreservation; the stronger role records at least two lawful nuisance-separated visible response-mode classes. The implication is operational. It determines neither the number of role-bearing blocks nor their factorization.

4.2 Factorization-relative role incidence

For $N \in \mathcal{N}_5(x)$ and $F \in \text{MaxFact}_5(N)$, let

$$\Omega_5(N, F) = \{\omega : \omega \text{ is a block of } F\}.$$

The role-standing signature of a block is

$$\begin{aligned} r \in \text{RSig}_5(\omega; N, F) \iff & \text{deleting } \omega \text{ at the fixed parent and parameter point,} \\ & \text{without compensatory repair, destroys the witness for } r \\ & \text{or a constitutive path required for that witness's standing.} \end{aligned} \quad (12)$$

Admissibility requires $\text{RSig}_5(\omega; N, F) \neq \emptyset$ for every block.

Set

$$I_{N,F} = \{(r, \omega) : r \in \text{RSig}_5(\omega; N, F)\} \subseteq R_5 \times \Omega_5(N, F).$$

An internal pattern is

$$P_{N,F} = (A_{N,F}, \preceq_{N,F}, q_{N,F}, \eta_{N,F}), \quad (13)$$

where $A_{N,F} = \{a_\omega : \omega \in \Omega_5(N, F)\}$,

$$q_{N,F}(r) = \{a_\omega : (r, \omega) \in I_{N,F}\},$$

and $\eta_{N,F}$ retains all constitutive edges and overlap data. Thus

$$\text{B}_P(a_\omega) = \text{RSig}_5(\omega; N, F). \quad (14)$$

Proposition 4.1 (Shared occupancy preserves role distinction). *If $|\Omega_5(N, F)| = 1$, the unique factor block is shared while r_Δ and $r_{\triangleright\triangleleft}$ remain distinct role projections.*

Proof. Deleting the unique core block destroys the complete ordered role packet, so its role-standing signature contains both roles. The roles retain different witness definitions and their strict order; sharing does not merge the predicates. $\square$

4.3 Why internal factors are not ROC roles

The address set $A_{N,F}$ indexes the role-bearing blocks composing one realization. It may vary with N and F . It is not substituted for the separately declared terminal role set R_5^{term} .

Theorem 4.2 (Two-level typing). *The internal and terminal data have different types:*

$$(A_{N,F}, \Omega_5(N, F), \sigma) \quad \text{and} \quad (R_5^{\text{term}}, \text{Residue}_{\text{term}}^{\text{live}}, \tau).$$

No internal assignment object may be identified with Σ_5^{term} without an explicit theorem connecting the domains and codomains.

Proof. $A_{N,F}$ indexes the complete factor blocks of one realization. R_5^{term} is fixed by the declared carrier-resolution target architecture before any residue is inspected. Their codomains are respectively one factorization and the carrier-semantic terminal live residue formed only after E_5^{term} and J_5^{term} are evaluated. $\square$

5 Witness support, semantic occupation, and the role-bearing core

5.1 Semantic occupation

A source element occupies a role only through a target-facing witness and its standing-bearing source ancestry. Mere occurrence in a Lagrangian, state space, dilation, or effective description is insufficient.

Definition 5.1 (Semantic role occupation). *Let $x \in \mathfrak{S}_v^{(4)}$ and let S be a finite source-coherent subdiagram of its PCR-4 witness architecture. Write $\text{Occupies}_5(S, r; x)$ when:*

1. *S contains or constitutively supports the complete witness for r ;*
2. *its source path begins at active carrier data of x and ends at the fixed interface;*
3. *deleting S 's contribution without repair destroys that witness or its standing;*
4. *the use preserves the PCR-4 parameter point, boundary, continuation, common-cone, and failure data.*

Lemma 5.2 (No response without transfer). *At the frozen target,*

$$\text{Occupies}_5(S, r_\Delta; x) \implies \text{Occupies}_5(\text{cl}_{\text{src}}^x(S), r_{\triangleright\Delta}; x).$$

Proof. The PCR-1 stronger role is defined on the common charged-current interface and contains nontrivial boundary transfer as a prerequisite. Source closure adds only the constitutive support required to expose that prerequisite. $\square$

5.2 Joint witness support

For each survivor, PCR-4 supplies finite witness components

$$W_{\text{role}}, W_{\text{chan}}, W_{\text{resp}}, W_{\text{rel}}, W_{\text{sim}} \subseteq \mathcal{W}_4(x).$$

Define

$$\text{JointSupp}_5(x) = \text{cl}_{\text{src}}^x(W_{\text{role}} \cup W_{\text{chan}} \cup W_{\text{resp}} \cup W_{\text{rel}} \cup W_{\text{sim}}). \quad (15)$$

Theorem 5.3 (Joint-support existence and role coverage). *For every $x \in \mathfrak{S}_v^{(4)}$, $\text{JointSupp}_5(x)$ is finite, source coherent, and contains standing-bearing witnesses for both roles, together with one PCR-4 physical common cone.*

Proof. This is the finite source closure of the witness packet imported from PCR-4. Finiteness follows from the finite-presentation comparison class. Source coherence and common-cone commutativity are PCR-4 survivor conditions. $\square$

What the theorem does not say

The common cone proves co-instantiation in one carrier architecture. It does not determine a unique normal form, a unique factorization, or a factor count.

5.3 The target-generated grammar

The grammar $\mathcal{G}_5$ is fixed before any survivor is inspected. Its sorts are generated from:

1. active source nodes and incidence maps from PCR-3;
2. the two PCR-1 role-witness sorts;
3. PCR-2 boundary, channel, response, continuation, and lawful-redescription sorts;
4. PCR-4 relation, common-cone, parameter, anomaly, vacuum, EFT, and failure sorts.

Its reductions may delete only target-inert material, remove presentation duplicates, contract a proved source-constitutive identity, replace an isomorphic cone presentation, or delete a source-null contribution. They may not erase a mandatory witness path or parent carrier coordinate.

Proposition 5.4 (Grammar independence). *The formation rules of $\mathcal{G}_5$, including the role-core and factorization predicates, do not depend on which PCR-4 family survives or on the desired endpoint cardinality.*

Proof. Every sort and equation is generated from the frozen target or an upstream witness type. No rule mentions a family name, benchmark orbit, field count, operator dimension, factor count, or endpoint branch. $\square$

5.4 Role-bearing core and auxiliary trace

For $N \in \mathcal{N}_5(x)$, let $\text{CorePath}_5(N)$ be the union of all nodes and edges occurring on at least one mandatory standing-bearing path for

$$W_{\text{role}}, W_{\text{chan}}, W_{\text{resp}}, W_{\text{rel}}, W_{\text{sim}}.$$

Define

$$\text{Core}_5(N) = \text{cl}_{\text{src}}^N(\text{CorePath}_5(N)), \quad \text{Aux}_5(N) = N \setminus \text{Core}_5(N), \quad (16)$$

where $\text{Aux}_5(N)$ is retained as typed trace data with every incidence edge back to the core.

Theorem 5.5 (Role-core existence and completeness). *For every $x \in \mathfrak{S}_v^{(4)}$ and $N \in \mathcal{N}_5(x)$:*

1. $\text{Core}_5(N) \neq \emptyset$;
2. $\text{Core}_5(N)$ is finite and source coherent;
3. the complete ordered role/interface witness diagram factors through $\text{Core}_5(N)$ and its retained constitutive maps;
4. no element of $\text{Aux}_5(N)$ is promoted to an occupancy block merely because it carries parent-carrier, parameter, continuation, or failure information.

Proof. Normal-form preservation retains both role witnesses and the complete interface/common-cone packet. Hence at least one mandatory witness path exists and its source closure is nonempty. Finiteness and coherence descend from N . By construction, every mandatory witness path lies in the core. Complementary trace data remain in the parent invariant but have no occupancy standing unless they occur on a mandatory role path. $\square$

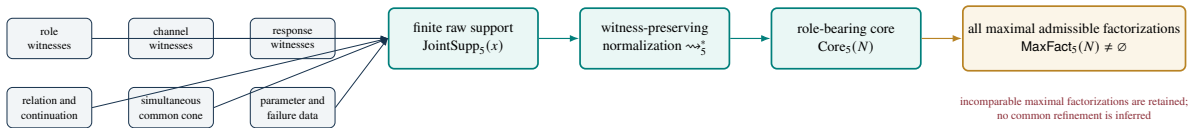


Figure 2: PCR-4 supplies one finite witness support. PCR-5 normalizes it, extracts the role-bearing core, and retains all maximal admissible factorizations rather than intersecting standing cuts.

6 Witness-preserving normalization and maximal role-bearing factorizations

6.1 Reduction relation

For a finite support diagram D generated by $\mathcal{G}_5$, write $D \rightsquigarrow_5 D'$ when one of the following target-governed moves is applied:

1. delete a nonempty source-convex subdiagram whose removal preserves every mandatory witness, every standing-bearing constitutive path, every continuation and failure record, and the complete parent carrier trace;

2. remove a duplicate witness presentation already identified by an imported lawful redescription;
3. contract an isomorphic source-constitutive chain whose internal vertex has no independent role incidence, boundary, continuation, parameter, carrier, or failure datum;
4. replace a witness cone by a fixed lower-complexity normal presentation of an isomorphic cone;
5. remove a source-null contribution at the fixed parameter point.

Every move is subject to preservation of the complete PCR-5 witness packet and full parent carrier trace.

Define the lexicographic measure

$$\mu(D) = (n_{\text{raw}}(D), c_{\text{pres}}(D)) \in \mathbb{N}^2,$$

where n_{raw} is support cardinality and c_{pres} is the fixed presentation complexity. Rules 1, 2, 3 and 5 reduce n_{raw} ; rule 4 preserves or reduces n_{raw} and strictly reduces c_{pres} .

Lemma 6.1 (Normalization termination). *Every $\rightsquigarrow_5$ -sequence terminates.*

Proof. Every permitted move strictly decreases μ , and lexicographic order on $\mathbb{N}^2$ is well founded. $\square$

Lemma 6.2 (Witness and carrier preservation). *If $D \rightsquigarrow_5 D'$, then D' preserves every mandatory role, channel, response, relation, common-cone, boundary, continuation, parameter, and failure witness of D , together with the complete parent carrier trace.*

Proof. Preservation is a side condition of every reduction rule. A primitive parent node may disappear from the displayed role-bearing support only through a proved presentation contraction; its parent identity remains in $\chi_{\text{car}}(x)$. $\square$

Definition 6.3 (Normal-form set).

$$\mathcal{N}_5(x) = \{N : \text{JointSupp}_5(x) \rightsquigarrow_5^* N, N \text{ is } \rightsquigarrow_5 \text{-irreducible}\}.$$

Confluence is not assumed. Distinct irreducible endpoints are retained.

Theorem 6.4 (Normal-form existence). *For every $x \in \mathfrak{S}_v^{(4)}$, $\mathcal{N}_5(x) \neq \emptyset$.*

Proof. Begin from finite $\text{JointSupp}_5(x)$ and repeatedly apply any available reduction. Termination produces an irreducible endpoint, while lemma 6.2 preserves the complete witness packet and parent trace. $\square$

6.2 Admissible role-bearing factorizations

Fix $N \in \mathcal{N}_5(x)$ and write $C = \text{Core}_5(N)$.

Definition 6.5 (Role-standing signature of a block). *For a nonempty source-convex block $B \subseteq C$, define $\text{RSig}_5(B; N) \subseteq R_5$ by declaring $r \in \text{RSig}_5(B; N)$ exactly when deleting B at the same parent carrier and parameter point, retaining no compensatory repair, destroys the witness for r or a constitutive path required for that witness's standing.*

Definition 6.6 (Admissible role-bearing factorization). *An admissible factorization is a pair*

$$F = (\mathcal{B}_F, \eta_F) \in \text{Fact}_5(N)$$

with $\mathcal{B}_F = \{B_1, \dots, B_k\}$ a finite partition of C and η_F the complete inter-block constitutive graph, satisfying:

1. **full coverage:** $C = \bigsqcup_i B_i$;
2. **source coherence:** every block is source-convex in the core graph;

3. **constitutive reconstruction:** the blocks and η_F reconstruct every mandatory role/interface witness and its standing;
4. **nonempty role standing:** $\text{RSig}_5(B_i; N) \neq \emptyset$ for every block;
5. **activity:** every block is non-null at the fixed PCR-3/PCR-4 point;
6. **no duplicates:** distinct blocks are not presentation duplicates;
7. **no hidden repair:** deletion loss cannot be repaired with another carrier, parameter point, or unregistered interaction;
8. **parent preservation:** $\chi_{\text{car}}(x)$ and $\text{Aux}_5(N)$ remain attached as complete invariants.

The refinement order is

$$F' \succeq F$$

when every block of F' is contained in a block of F , every edge of η_F is reconstructed by $\eta_{F'}$, and F' remains admissible. Thus larger elements are finer factorizations.

Lemma 6.7 (One-block admissibility). *The factorization*

$$F_{\text{coarse}} = (\{\text{Core}_5(N)\}, \eta_{\text{int}})$$

is admissible. Hence $\text{Fact}_5(N) \neq \emptyset$.

Proof. The role-bearing core is finite, source coherent, and reconstructs the complete witness packet by theorem 5.5. Deleting the whole core destroys both role witnesses and their standing, so its role-standing signature is nonempty. Activity, no-repair, and parent preservation are inherited from the fixed survivor point. There are no distinct blocks to duplicate. $\square$

Theorem 6.8 (Maximal admissible factorization existence). *For every $x \in \mathfrak{S}_v^{(4)}$ and $N \in \mathcal{N}_5(x)$,*

$$\text{MaxFact}_5(N) = \{F \in \text{Fact}_5(N) : F \text{ has no proper admissible refinement}\} \neq \emptyset.$$

Proof. The finite core has a finite partition lattice, so $\text{Fact}_5(N)$ is a finite nonempty poset. Every finite nonempty poset has a maximal element. $\square$

Definition 6.9 (Factorization-relative occupation cells). *For $F \in \text{MaxFact}_5(N)$, define*

$$\Omega_5(N, F) = \mathcal{B}_F. \tag{17}$$

These blocks are the occupancy cells of the selected maximal factorization. Several incomparable maximal factorizations are all retained.

Theorem 6.10 (Role-bearing-cell theorem). *For every $F \in \text{MaxFact}_5(N)$ and every $\omega \in \Omega_5(N, F)$,*

$$\text{RSig}_5(\omega; N) \neq \emptyset.$$

No auxiliary-only trace object is an occupancy cell.

Proof. Nonempty role standing is clause 4 of admissible factorization. Maximality does not create the property; it selects a maximally refined factorization from objects already satisfying it. Auxiliary data are excluded from the core unless they lie on a mandatory role-standing path. $\square$

Corollary 6.11 (No common-refinement inference). *The analysis never intersects admissible factorizations. Incomparable maximal factorizations remain distinct internal possibilities unless the lawful groupoid proves them isomorphic.*

Corollary 6.12 (No forced one-factor conclusion). *PCR-4 common-cone coherence is compatible with one block, several nested blocks, several distributed blocks, and several incomparable maximal factorizations. Factorization cardinality is a PCR-5 output.*

7 Pattern derivation from maximal factorization graphs

7.1 Factor graph and role incidence

Fix $x \in \mathfrak{S}_v^{(4)}$, $N \in \mathcal{N}_5(x)$, and $F \in \text{MaxFact}_5(N)$. Let $G_{N,F}$ be the finite directed graph whose vertices are the blocks of $\Omega_5(N, F)$ and whose edges are the retained constitutive maps in η_F . Define

$$I_{N,F} = \{(r, \omega) : r \in \text{RSig}_5(\omega; N)\}.$$

Every vertex has nonempty role incidence by theorem 6.10. The pattern is a function of $(G_{N,F}, I_{N,F})$, not a free input.

7.2 Three factorization-relative pattern types

Definition 7.1 (Shared pattern). $P_{N,F} = \mathbf{P}_{\text{sh}}$ iff $|\Omega_5(N, F)| = 1$. *The unique block has a role-standing signature containing both roles.*

Definition 7.2 (Nested-distinct pattern). $P_{N,F} = \mathbf{P}_{\text{nd}}$ iff $|\Omega_5(N, F)| > 1$, *the graph contains distinct minimal role kernels for $r_{\bowtie}$ and r_{Δ} , and the stronger-role support contains a directed constitutive path to the weaker kernel, while no mandatory witness standing is borne only by a genuinely branching joint subdiagram.*

Definition 7.3 (Distributed pattern). $P_{N,F} = \mathbf{P}_{\text{dist}}$ iff $|\Omega_5(N, F)| > 1$ *and the factor graph is not nested-distinct. Equivalently, at least one mandatory role witness or its standing is borne only by a jointly necessary finite subdiagram rather than by one nested role kernel.*

The distributed case is the complement of the nested-distinct case among multi-block admissible factorizations. It does not require a prescribed number of blocks.

7.3 Address realization

For every (N, F) , use the complete address set

$$A_{N,F} = \{a_\omega : \omega \in \Omega_5(N, F)\}. \quad (18)$$

The role-incidence map is

$$q_{N,F}(r) = \{a_\omega : (r, \omega) \in I_{N,F}\},$$

and $\eta_{N,F}$ is the complete edge and overlap data of $G_{N,F}$. No address or factor block is optional.

Lemma 7.4 (Factorization-isomorphism invariance). *A lawful isomorphism between (N, F) and (N', F') preserves the factor graph, role-standing signatures, and pattern type.*

Proof. A lawful isomorphism preserves block traces, deletion behavior, constitutive edges, and mandatory witness paths. It therefore preserves $(G_{N,F}, I_{N,F})$ and the defining case distinction. $\square$

Pattern is calculated, not selected

A common cone does not force the shared pattern. A maximal factorization can be shared, nested-distinct, or distributed, and one normal form may admit several incomparable maximal factorizations. Every such factorization remains live until the lawful quotient is applied.

8 Factorization and pattern completeness

Theorem 8.1 (Pattern trichotomy). *For every $x \in \mathfrak{S}_v^{(4)}$, every $N \in \mathcal{N}_5(x)$, and every $F \in \text{MaxFact}_5(N)$, exactly one of*

$$P_{\text{sh}}, \quad P_{\text{nd}}, \quad P_{\text{dist}}$$

is induced by $(G_{N,F}, I_{N,F})$.

Proof. If the factorization has one block, it is shared. Assume it has more than one. If the distinct role kernels and directed constitutive relation satisfy the nested-distinct definition, that case holds. Otherwise the factorization is distributed by definition. The cases are disjoint by block cardinality or graph condition and exhaustive by the finite case split. $\square$

Corollary 8.2 (Complete address cardinality). *For every maximal factorization,*

$$|A_{N,F}| = |\Omega_5(N, F)|,$$

and the tautological map $a_\omega \mapsto \omega$ is a bijection onto the entire factorization.

Proof. $A_{N,F}$ is a labeled copy of all factor blocks. $\square$

8.1 Excluded factorization forms

The grammar rejects:

1. selecting a pattern before normalization and maximal factorization;
2. using a proper subset of $\Omega_5(N, F)$ as the assignment codomain;
3. promoting auxiliary-only carrier or failure data to occupancy blocks;
4. accepting a block with empty role-standing signature;
5. forcing all common-cone supports into one block;
6. intersecting incomparable admissible factorizations without a new admissibility proof;
7. discarding an incomparable maximal factorization because it is inconvenient;
8. splitting a source-constitutive block into presentation duplicates;
9. omitting an edge required by a role witness or its standing.

Theorem 8.3 (Factorization-relative pattern completeness). *Relative to $\mathcal{G}_5$, every finite internal occupancy architecture has at least one complete maximal-factorization presentation and every such presentation has exactly one of the three pattern types.*

Proof. Normal-form existence supplies N . Role-core existence supplies a finite nonempty core. The one-block factorization and finite-poset maximality supply at least one $F \in \text{MaxFact}_5(N)$. Theorem 8.1 supplies the pattern, and corollary 8.2 supplies complete address coverage. The claim is relative to the declared grammar and does not assert uniqueness of F or exclusion of future target-generated pattern sorts. $\square$

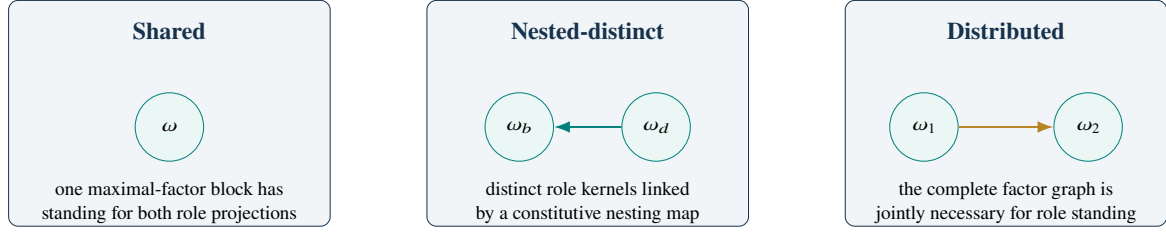


Figure 3: Pattern types are read from one maximal admissible factorization. Several incomparable maximal factorizations may coexist and are never replaced by an unproved common refinement.

9 Pointwise eligibility on maximal role-bearing factorizations

9.1 Eligibility domain

Fix $x \in \mathfrak{S}_v^{(4)}$, $N \in \mathcal{N}_5(x)$, $F \in \text{MaxFact}_5(N)$, a block $\omega \in \Omega_5(N, F)$, and its address $a_\omega \in A_{N,F}$. The role bundle is fixed by $\text{RSig}_5(\omega; N)$ before eligibility is evaluated.

Definition 9.1 (PCR–5 eligibility predicate).

$$E_5(x, N, F, \omega, a_\omega) = 1 \quad (19)$$

iff all of the following hold at one source parameter point:

1. **role-standing adequacy:** $\text{RSig}_5(\omega; N) \neq \emptyset$, and the block realizes or constitutively supports every role recorded in that signature;
2. **positive witness:** the contribution factors through the preserved PCR–4 witness packet;
3. **deletion sensitivity:** deleting the block without repair destroys an assigned role witness or a standing-bearing constitutive path needed for it;
4. **source activity:** the contribution is non-null after PCR–3 pointwise activity reduction;
5. **target participation:** the contribution lies on a mandatory source-to-interface role path, directly or constitutively;
6. **derived-factor typing:** block, address, role incidence, and constitutive edge type agree in $(G_{N,F}, I_{N,F})$;
7. **trace sufficiency:** every grammar coordinate required to evaluate the block is defined;
8. **one-survivor standing:** no witness coordinate is borrowed from another carrier point;
9. **lawful invariance:** the value is unchanged by $\mathcal{G}_5$ -lawful presentation isomorphism.

A source block can be eligible even when its observable role projection is mediated by another block: it must be deletion-sensitive for the standing of the complete role witness. Purely auxiliary data remain in $\text{Aux}_5(N)$ and are not evaluated as occupancy blocks.

9.2 Eligibility classification

Each typed block-address pair receives

$$\text{Disp}_E(x, N, F, \omega, a_\omega) \in \{\text{PASS}, \text{FAIL}, \text{N/A}, \text{TARGET UPDATE}\}.$$

N/A is never a pass. Missing a mandatory grammar coordinate fails closed; TARGET UPDATE is reserved for a proposed use that changes the target or source class.

Proposition 9.2 (No inapplicability laundering). *A mandatory role-standing, witness, or constitutive coordinate cannot be omitted and reclassified as N/A.*

Proof. The core, factorization, and grammar sorts are fixed before the disposition is computed. Absence of an activated coordinate is therefore a failure of typing or trace sufficiency. $\square$

9.3 Eligibility transfer

Theorem 9.3 (Maximal-factorization eligibility transfer). *For every $x \in \mathfrak{S}_v^{(4)}$, every $N \in \mathcal{N}_5(x)$, every $F \in \text{MaxFact}_5(N)$, and every $\omega \in \Omega_5(N, F)$,*

$$E_5(x, N, F, \omega, a_\omega) = 1.$$

Proof. Admissible factorization gives nonempty role-standing signature, deletion sensitivity, activity, full core coverage, no hidden repair, and parent preservation. Normalization preserves the complete PCR-4 witness packet. Core membership gives target participation. The address is derived from the block itself, giving exact factor typing. Every retained coordinate belongs to one survivor and one parameter point, and all clauses are invariant under lawful grammar isomorphisms. Thus every block passes. $\square$

Eligibility is not comparative merit

Eligibility does not rank operator dimension, field count, simplicity, naturalness, ultraviolet completion, or phenomenological familiarity. Such coordinates have no selector standing under the frozen target.

10 Eligibility completeness relative to the fixed grammar

10.1 Independent generation of eligibility coordinates

The primitive coordinate families are generated independently from the frozen role, source, interface, core, and factorization sorts:

$$\begin{aligned} E_{\mathcal{G}_5} = & \{\text{RoleStanding}, \text{Witness}, \text{Delete}, \text{Active}\} \\ & \cup \{\text{TargetPath}, \text{FactorType}, \text{Boundary}, \text{Continuation}\} \\ & \cup \{\text{Parameter}, \text{Failure}, \text{Ownership}, \text{Redescription}\}. \end{aligned}$$

No family name, benchmark output, factor count, or endpoint branch occurs in this list.

Definition 10.1 (Grammar-admissible eligibility use). *A grammar-admissible eligibility use is a same-target functional generated by one or more coordinate families in $E_{\mathcal{G}_5}$, evaluated on a typed maximal-factorization block and invariant under imported lawful redescrptions. Let $\text{EligUse}_{\mathcal{G}_5}$ denote the class generated by finite composition and finite products of these primitive uses.*

Lemma 10.2 (Eligibility-coordinate descent). *Every $U_E \in \text{EligUse}_{\mathcal{G}_5}$ factors through*

$$(\tau_{\text{occ}}^{(5)}(\omega; x), \text{RSig}_5(\omega; N), G_{N,F}, I_{N,F}, \chi_{\text{car}}(x)).$$

Proof. The primitive generators are exactly fields of the displayed tuple, and factorization is preserved by finite composition and products. This is a generation theorem from a fixed grammar, not a trace-extensionality assumption. $\square$

Theorem 10.3 (Grammar-relative eligibility completeness). *For every typed block-address pair generated by $\mathcal{G}_5$, the disposition is total and*

$$E_5 = 1 \iff U_E \text{ is positive for every } U_E \in \text{EligUse}_{\mathcal{G}_5}. \quad (20)$$

Proof. Totality follows because every generated primitive coordinate is defined or fails closed on a typed grammar object. The forward implication follows clause by clause from definition 9.1. Conversely, each

clause of definition 9.1 is a primitive grammar-admissible use. If $E_5 = 0$, the failed clause supplies a negative use. $\square$

10.2 Scope of eligibility completeness

The theorem is complete relative to $\mathcal{G}_5$ and the frozen target. A future independently forced observable may enlarge the grammar. The paper does not infer that no such coordinate can exist; it proves that no coordinate generated by the declared PCR–5 grammar is silently omitted.

11 Complete-factorization compatibility and transfer

11.1 Compatibility is higher arity

Let

$$z = (x, N, F, P_{N,F}, \sigma, e)$$

be a full-factor, pointwise eligible internal preassignment. Compatibility is a predicate on the entire factorized diagram, not a product of pairwise block tests.

Definition 11.1 (PCR–5 compatibility predicate). $J_5(z) = 1$ iff:

1. every address and every factor block is used exactly once;
2. all blocks belong to one parent survivor and one source parameter point;
3. every edge and overlap in η_F is realized by the corresponding source map;
4. the inherited relation $r_\Delta \succ r_{\triangleright\triangleleft}$ holds on the complete diagram;
5. event-channel and visible-response witnesses are the PCR–4 common witnesses;
6. boundary, continuation, representation, charge, anomaly, vacuum, and EFT data agree across the diagram;
7. no support is duplicated and no failed coordinate is repaired by another block;
8. the complete witness packet admits the preserved PCR–4 physical common cone;
9. every imported failure condition remains failed closed;
10. every auxiliary trace coordinate is attached to its correct parent/core incidence and is not counted as an additional occupancy block.

11.2 Compatibility transfer

Theorem 11.2 (Common-cone compatibility transfer). Let $x \in \mathfrak{S}_v^{(4)}$, $N \in \mathcal{N}_5(x)$, $F \in \text{MaxFact}_5(N)$, and let $\sigma_{N,F} : a_\omega \mapsto \omega$ be the full-factor address bijection. Then

$$J_5(x, N, F, P_{N,F}, \sigma_{N,F}) = 1.$$

Proof. PCR–4 proves that all role, channel, response, relation, and simultaneous witnesses belong to one carrier architecture, one parameter point, and one physical common cone. Normalization preserves those witnesses and maps. Role-core extraction preserves every mandatory witness path. Factorization admissibility reconstructs the whole core from its blocks and retained edges, while the full-factor bijection uses every block exactly once. Auxiliary data remain attached to the same parent. Thus the complete factorized diagram inherits all PCR–4 coherence and failure conditions. The argument transfers compatibility; it does not infer a factor count or unique factorization. $\square$

11.3 Grammar-relative compatibility completeness

Let $J_{\mathcal{G}_5}$ be the primitive compatibility uses generated by full coverage, ownership, source incidence, role nesting, channel-response identity, boundary/continuation identity, common consistency ledgers, parameter identity, nonduplication, no repair, common-cone existence, failure preservation, and auxiliary/core typing.

Theorem 11.3 (Grammar-relative compatibility completeness). *For every full-factor pointwise eligible internal assignment candidate,*

$$J_5(z) = 1 \iff U_J(z) \text{ is positive for every } U_J \in J_{\mathcal{G}_5}. \quad (21)$$

Proof. Every clause of definition 11.1 is a primitive generated compatibility use, so a failed clause supplies a negative use. Conversely, the conjunction fixes every compatibility coordinate generated by the frozen witness/factorization grammar. Finite composition cannot create a new negative value when all primitive values and incidences are positive. $\square$

Proposition 11.4 (Pairwise passes are insufficient). *Pairwise compatible block traces do not imply $J_5 = 1$.*

Proof. A family of pairwise compatible blocks can still lack one common parameter point, one global anomaly/vacuum/EFT ledger, one core reconstruction, or one common cone. Those are higher-arity clauses. $\square$

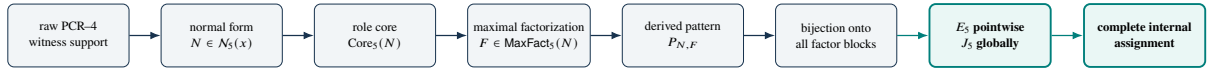


Figure 4: Internal analysis pipeline: normalize, extract the role core, retain a maximal admissible factorization, derive its pattern, cover every factor block, then evaluate E_5 and J_5 .

12 One-survivor ownership and cross-survivor exclusion

Definition 12.1 (One-survivor ownership). *An internal assignment has one-survivor ownership when every normal-form node, role-core block, auxiliary trace, witness, parameter, boundary record, continuation map, constitutive edge, and failure condition is a restriction of one parent $x \in \mathfrak{S}_v^{(4)}$.*

Theorem 12.2 (No cross-survivor assignment). *No element of Σ_5^{int} can contain factor blocks or auxiliary traces from carrier-inequivalent parents.*

Proof. The raw support begins inside one $\mathcal{W}_4(x)$, every normalization move stays inside that source diagram, role-core extraction and factorization occur inside the same normal form, and J_5 includes one-survivor ownership. A cross-parent splice lacks one common source point and one PCR-4 common cone. $\square$

Proposition 12.3 (Matching does not transfer ownership). *An EFT matching map between two carrier architectures cannot make blocks or traces of one architecture owned by the other.*

Proof. Matching may preserve a low-energy coefficient or interface observable while forgetting primitive nodes, source incidence, continuation, and failure domains. Ownership is a parent-carrier relation and is not transported by a noninvertible projection. $\square$

The world-indexed assignment type is

$$\sigma_x : A_{N,F} \xrightarrow{\sim} \Omega_5(N, F), \quad N \in \mathcal{N}_5(x), \quad F \in \text{MaxFact}_5(N). \quad (22)$$

No assignment is assembled by choosing an r_{pcr} -block from one survivor and an r_{Δ} -block from another.

13 Complete internal assignments and universal nonempty fibers

13.1 Internal liveness is factorization relative

For fixed (x, N, F) , the complete internal live-factor set is

$$\text{OccCell}_5^{\text{live}}(x; N, F) := \Omega_5(N, F).$$

No proper subset is permitted as an assignment codomain.

Definition 13.1 (Internal complete assignment over a maximal factorization). *For $N \in \mathcal{N}_5(x)$ and $F \in \text{MaxFact}_5(N)$, an internal complete assignment is*

$$z = (x, N, F, P_{N,F}, \sigma, e, j),$$

where

$$\sigma : A_{N,F} \xrightarrow{\sim} \Omega_5(N, F),$$

all block-address pairs pass E_5 , and $J_5(z) = 1$. Let $\Sigma_5^{\text{int}}(x; N, F)$ be the set of such assignments.

Define the survivor assignment fiber before quotient by

$$\Sigma_5^{\text{int}}(x) = \bigsqcup_{N \in \mathcal{N}_5(x)} \bigsqcup_{F \in \text{MaxFact}_5(N)} \Sigma_5^{\text{int}}(x; N, F). \quad (23)$$

A survivor is internally live when this set is nonempty.

13.2 Universal nonempty-fiber theorem

Theorem 13.2 (Every PCR-4 survivor has a complete internal assignment). *For every $x \in \mathfrak{S}_v^{(4)}$,*

$$\Sigma_5^{\text{int}}(x) \neq \emptyset. \quad (24)$$

Consequently the internally live survivor residue equals $\mathfrak{S}_v^{(4)}$.

Proof. Choose $N \in \mathcal{N}_5(x)$, which exists by theorem 6.4. Its role-bearing core is nonempty by theorem 5.5. Choose $F \in \text{MaxFact}_5(N)$, which exists by theorem 6.8. The pattern and complete address set are derived by theorem 8.1 and corollary 8.2. The tautological full-factor bijection passes every eligibility clause by theorem 9.3 and complete compatibility by theorem 11.2. Hence it defines an element of $\Sigma_5^{\text{int}}(x; N, F)$. $\square$

Remark 13.3 (No canonical physical section). *The proof establishes nonemptiness. It does not assert normal-form confluence, unique maximal factorization, a preferred assignment, or one orbit per carrier fiber. Incomparable maximal factorizations remain available to the quotient.*

13.3 Separation from terminal carrier residue

The terminal ROC live residue is constructed later from a separately defined carrier-semantic candidate set, terminal eligibility relation, and terminal compatibility predicate. The factorization-sensitive internal object $\mathfrak{D}_5^{\text{int}}$ is first projected by $\bar{\pi}_{\text{car}}$; maximal-factorization variants over one frozen parent therefore do not automatically become distinct terminal residues. Internal factor blocks compose an occupancy architecture, while terminal residues are carrier-semantic alternatives.

14 The complete internal assignment space

Definition 14.1 (Internal assignment space).

$$\Sigma_5^{\text{int}} := \bigsqcup_{x \in \mathfrak{S}_V^{(4)}} \bigsqcup_{N \in \mathcal{N}_5(x)} \bigsqcup_{F \in \text{MaxFact}_5(N)} \Sigma_5^{\text{int}}(x; N, F). \quad (25)$$

Every element contains the complete parent, normal form, role core, auxiliary trace, maximal factorization, derived pattern, full-factor bijection, eligibility certificate, and compatibility certificate.

14.1 No subset codomain

The codomain condition is literal:

$$\text{im}(\sigma) = \Omega_5(N, F). \quad (26)$$

A map onto $C \subsetneq \Omega_5(N, F)$ is not an internal complete assignment.

Proposition 14.2 (Unused-factor rejection). *If $\omega \in \Omega_5(N, F) \setminus \text{im}(\sigma)$, then $\sigma \notin \Sigma_5^{\text{int}}(x; N, F)$.*

Proof. Every factor block has nonempty role-standing signature. Omitting it leaves part of a role witness or its constitutive standing unassigned and violates both full coverage and the first clause of J_5 . $\square$

14.2 Assignment completeness

Theorem 14.3 (Complete internal assignment-space theorem). *Relative to $\mathcal{G}_5$, Σ_5^{int} contains every and only finite one-survivor assignments over maximal admissible role-bearing factorizations satisfying the derived pattern, E_5 , and J_5 .*

Proof. Every grammar-generated finite support has a normal form, nonempty role core, at least one maximal admissible factorization, complete factor address set, and derived pattern. Every complete assignment over that object is included by the disjoint-union definition. Conversely, every included tuple is grammar generated, covers the full factorization, and carries positive eligibility and compatibility certificates. No model-name enumeration or common-refinement construction is used. $\square$

14.3 Possible multiplicity inside one carrier fiber

One survivor may admit several nonisomorphic normal forms, several incomparable maximal factorizations, or several role-incidence-preserving assignment symmetries. Thus

$$|\Sigma_5^{\text{int}}(x)| \geq 1$$

does not imply one internal orbit over x . This possible additional multiplicity is retained and is one reason the endpoint is a lower bound rather than an exact two-orbit theorem.

15 The lawful internal assignment groupoid

15.1 Factorization-sensitive invariant object

For $z = (x, N, F, P_{N,F}, \sigma, e, j)$, define

$$\mathcal{I}_5(z) = (T_{V,\text{int}}^{(5)}, \mathcal{G}_5, \chi_{\text{car}}(x), N, \text{Core}_5(N), \text{Aux}_5(N), F, \Omega_5(N, F), G_{N,F}, I_{N,F}, P_{N,F}, \sigma, e, j, \tau_{\text{asg}}^{(5)}(z)). \quad (27)$$

Definition 15.1 (Lawful internal assignment groupoid). Γ_5^{int} has object space Σ_5^{int} . A morphism $z \rightarrow z'$ is an isomorphism of $\mathcal{I}_5(z)$ and $\mathcal{I}_5(z')$ preserving:

1. the internal target and grammar sorts;
2. the complete parent carrier trace;

3. normal form, role core, auxiliary attachments, and the selected maximal-factorization graph up to lawful presentation isomorphism;
4. every factor trace, role-standing signature, boundary, continuation, parameter, common-cone, and failure coordinate;
5. E_5 , J_5 , and the full-factor condition.

Allowed generators include factor-address relabeling, lawful basis/gauge/source-diagram redescription, lawful reparameterization within one stratum, isomorphic normal-form presentation, isomorphic maximal-factorization presentation, and contraction/expansion of proved presentation duplicates. Forbidden generators include interface-only identification, EFT matching as identity, cross-survivor splicing, factor deletion, intersection of incomparable factorizations, or endpoint-driven merge.

Theorem 15.2 (Maximal factorization-sensitive presentation quotient). *Within the declared internal grammar, Γ_5^{int} contains every invariant-preserving presentation isomorphism and no morphism that erases a generated carrier, role-standing, factorization, continuation, or failure coordinate.*

Proof. All invariant-preserving isomorphisms are admitted by definition. A larger factorization-sensitive presentation relation would identify nonisomorphic invariant objects and erase at least one generated coordinate. This theorem classifies the internal factorization-sensitive problem only; it does not assert that incomparable factorizations over one parent carrier are distinct at the coarser terminal carrier resolution. $\square$

Definition 15.3 (Internal factorization-sensitive orbit object).

$$\mathfrak{D}_5^{\text{int}} := \Sigma_5^{\text{int}} / \Gamma_5^{\text{int}}. \quad (28)$$

15.2 Carrier projection

Define

$$\pi_{\text{car}} : \Sigma_5^{\text{int}} \rightarrow \mathfrak{S}_v^{(4)}, \quad \pi_{\text{car}}(x, N, F, P_{N,F}, \sigma, e, j) = x. \quad (29)$$

Lemma 15.4 (Carrier projection is orbit constant). *π_{car} is constant on Γ_5^{int} -orbits up to the frozen PCR-3/PCR-4 carrier equivalence.*

Proof. Parent carrier trace is preserved by every internal groupoid morphism. $\square$

Hence there is a well-defined map

$$\bar{\pi}_{\text{car}} : \mathfrak{D}_5^{\text{int}} \rightarrow \mathfrak{S}_v^{(4)}. \quad (30)$$

No canonical or preferred section is claimed.

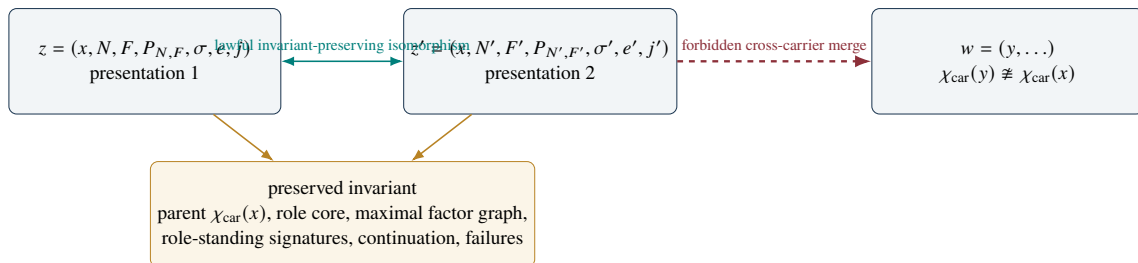


Figure 5: The internal groupoid removes factorization-sensitive presentation redundancy while preserving parent carrier trace. A separate terminal projection later forgets factorization decomposition at carrier resolution.

16 Factorization-sensitive internal quotient and carrier-resolution descent

16.1 Grammar-generated internal uses

Let $\text{AssignUse}_{\mathcal{G}_5}$ be the finite-composition closure of the primitive target, carrier, normal-form, role-core, auxiliary-trace, maximal-factorization, block-incidence, eligibility, compatibility, boundary, continuation, parameter, common-cone, and failure uses generated by $\mathcal{G}_5$.

Definition 16.1 (Grammar-semantic internal equivalence). *For $z, z' \in \Sigma_5^{\text{int}}$, write $z \equiv_{\text{asg}}^{\mathcal{G}_5} z'$ iff every $U \in \text{AssignUse}_{\mathcal{G}_5}$ has equal value on z, z' , up to the lawful presentation relation of its codomain.*

Lemma 16.2 (Generated-trace characterization).

$$z \equiv_{\text{asg}}^{\mathcal{G}_5} z' \iff I_5(z) \cong I_5(z') \quad \text{by a } \mathcal{G}_5\text{-lawful factorization-sensitive isomorphism.}$$

Proof. Every field of I_5 is a primitive generated internal use, while every generated internal use factors through those fields. The statement is therefore exact relative to the declared factorization-sensitive internal grammar. $\square$

Theorem 16.3 (Exact factorization-sensitive internal orbit theorem relative to $\mathcal{G}_5$). *For $z, z' \in \Sigma_5^{\text{int}}$,*

$$z \equiv_{\text{asg}}^{\mathcal{G}_5} z' \iff \text{Mor}_{\Gamma_5^{\text{int}}}(z, z') \neq \emptyset. \quad (31)$$

This is an exact presentation quotient for the declared internal decomposition problem. It is not asserted to be the terminal quotient of carrier realizations.

Proof. Combine lemma 16.2 with definition 15.1 and theorem 15.2. $\square$

16.2 Carrier-resolution equivalence forgets factorization presentation

The terminal target asks for the carrier-semantic realization class, not for a preferred maximal factorization. Define on $\mathfrak{D}_5^{\text{int}}$

$$q \equiv_{\text{term}} q' \iff \bar{\pi}_{\text{car}}(q) = \bar{\pi}_{\text{car}}(q') \quad \text{in } \mathfrak{S}_v^{(4)}. \quad (32)$$

Theorem 16.4 (Factorization-insensitive carrier-resolution quotient). *The quotient by (32) is canonically the frozen carrier-semantic survivor quotient:*

$$\boxed{\mathfrak{D}_5^{\text{int}} / \equiv_{\text{term}} \cong \text{im}(\bar{\pi}_{\text{car}}) = \mathfrak{S}_{v, \text{car}}^{(5)}} \quad (33)$$

Once universal nonempty carrier fibers are proved in section 21, $\mathfrak{S}_{v, \text{car}}^{(5)} = \mathfrak{S}_v^{(4)}$.

Proof. Equation (32) is the kernel equivalence of the carrier projection. Quotienting a set by the fibers of a map yields its image. The target resolution is therefore insensitive to alternative maximal factorizations over the same frozen carrier class. Conversely, carrier-inequivalent parents cannot be identified because $\bar{\pi}_{\text{car}}$ takes different values on them. $\square$

Corollary 16.5 (No factorization-overcount in the terminal residue). *Several incomparable maximal factorizations over one carrier may produce several internal factorization-sensitive orbits, but they contribute only one terminal carrier-semantic residue unless the frozen carrier trace itself differs.*

16.3 The lower bound needs less than internal exactness

The strict carrier-multiplicity result requires only valid complete internal assignments and preservation of parent carrier trace. In particular, even a future coarsening of factorization-sensitive internal structure cannot identify the law-only and active-three-singlet parents without a lawful carrier-trace isomorphism.

Proposition 16.6 (Two carrier-distinct witnesses suffice). *If $z_1, z_2 \in \Sigma_5^{\text{int}}$ and*

$$\chi_{\text{car}}(\pi_{\text{car}} z_1) \not\equiv \chi_{\text{car}}(\pi_{\text{car}} z_2),$$

then their terminal carrier-semantic residues are distinct and $|\mathfrak{S}_{\nu, \text{car}}^{(5)}| > 1$.

Proof. Carrier-distinct assignments lie in different fibers of $\bar{\pi}_{\text{car}}$, hence in different classes of $\equiv_{\text{term}}$. $\square$

Corollary 16.7 (No interface-only quotient). *The PCR–2 interface profile is too coarse to define the carrier-resolution realization quotient.*

Proof. Equal interface data can coexist with unequal frozen parent carrier traces. The latter are exactly the distinctions preserved by $\mathfrak{S}_{\nu, \text{car}}^{(5)}$. $\square$

17 Carrier-family factorization atlas

17.1 Family names do not determine factorization or pattern

PCR–3 and PCR–4 organize the candidate and survivor universes into source families. PCR–5 does not assign one factorization or occupancy pattern to a family name. Both are calculated pointwise from the normalized role-bearing core. The atlas therefore records nonempty assignment fibers over surviving loci, not family-wide block counts.

Table 2: PCR–4 family status and PCR–5 factorization disposition.

Family	PCR–4 status	PCR–5 theorem	Endpoint use
BASE	Strict base locus empty; overlaps may survive.	Every surviving overlap has a normal form, nonempty role core, and at least one maximal admissible factorization.	No independent lower-bound witness.
LAW	Nonempty exact locus.	Nonempty factorization fibers; the Weinberg benchmark supplies an explicit complete assignment without a fixed block-count claim.	Supplies q_W .
LOCAL/HID	Nonempty active loci.	Nonempty factorization fibers; the active three-singlet benchmark supplies an explicit complete assignment.	Supplies q_{3N} .
COMP	Implicit exact moduli locus.	Every surviving point has at least one maximal factorization and complete assignment.	May add further orbits.
TOP/EXT/BDY	Pure loci empty; hybrid survivors only.	The theorem applies to each surviving hybrid point without reviving excluded pure loci.	May add hybrid fibers only.
SOL	Implicit exact moduli locus.	Every surviving solitonic point has a nonempty factorization fiber.	Exact orbit count open.

Family	PCR–4 status	PCR–5 theorem	Endpoint use
ENV	Protocol-only environment excluded; source-bearing hybrids may survive.	Only source-standing survivors enter; each has a nonempty factorization fiber.	No protocol-only orbit.
HYB	Universal finite-hybrid survivor locus subject to PCR–4.	Full witness support normalizes, its role core factors maximally, and a pattern is derived for every surviving point.	Possible additional multiplicity.

Theorem 17.1 (Familywise nonempty-fiber transfer). *For every family $\mathcal{F}$ and every $x \in \mathfrak{S}_v^{(4)} \cap \mathcal{F}$,*

$$\Sigma_5^{\text{int}}(x) \neq \emptyset.$$

No PCR–4 empty pure locus is repopulated.

Proof. The first claim is theorem 13.2 restricted to a family. The second follows because the assignment domain is indexed by $x \in \mathfrak{S}_v^{(4)}$; an excluded PCR–4 point has no support to normalize or factor. $\square$

17.2 Factorization diversity is retained

A family can contain shared, nested-distinct, and distributed maximal factorizations, and one carrier point can contain several incomparable maximal factorizations. The atlas is therefore not an orbit count. It certifies only the universal nonempty-fiber statement needed for the carrier-indexed lower bound.

18 Explicit complete assignment for the law-only Weinberg benchmark

18.1 The PCR–4 law-only point

Choose a PCR–4 survivor point x_W in the nonempty law-only locus with nonzero complex symmetric coefficient κ_* for

$$O_5 = (LH)(LH).$$

Its parent trace contains the frozen Standard Model source/interface nodes and the active dimension-five law object, but no added primitive local neutral-fermion node.

For concreteness, choose a Takagi form

$$\kappa_* = UDU^T$$

with nontrivial mixing matrix U and positive diagonal D having the target-required response-mode separation. This fixes one point rather than a family-name surrogate.

18.2 Normal form, role core, and maximal factorization

Let D_W be the source closure of the active law object $\kappa_* O_5$, its constitutive L, H incidences, the induced light-neutrino propagation kernel, and the complete PCR–4 role, channel, response, relation, continuation, and common-cone packet.

Choose any

$$N_W \in \mathcal{N}_5(x_W).$$

By theorem 5.5, $\text{Core}_5(N_W)$ is nonempty and reconstructs the full witness packet. The one-block factorization

is admissible, so by theorem 6.8 there exists at least one

$$F_W \in \text{MaxFact}_5(N_W).$$

Every block of F_W has nonempty role-standing signature. The proof neither requires nor asserts that F_W is unique or one-block.

Theorem 18.1 (Weinberg benchmark complete assignment). *Let $P_W = P_{N_W, F_W}$, let $A_W = A_{N_W, F_W}$, and let*

$$\sigma_W : A_W \xrightarrow{\sim} \Omega_5(N_W, F_W)$$

be the tautological full-factor bijection. Then

$$z_W = (x_W, N_W, F_W, P_W, \sigma_W, e_W, j_W) \in \Sigma_5^{\text{int}}.$$

Proof. Factorization admissibility and theorem 9.3 give $E_5 = 1$ on every block. One parent, one parameter point, complete core reconstruction, retained inter-block maps, one PCR–4 common cone, and unchanged gauge, anomaly, vacuum, EFT, continuation, and failure ledgers give $J_5 = 1$ by theorem 11.2. The assignment covers every factor block. $\square$

Define

$$q_W = [z_W]_{\Gamma_5^{\text{int}}} \in \mathfrak{D}_5^{\text{int}}.$$

Its complete invariant records no added primitive local neutral-fermion node. Any additional maximal factorization of the same normal form remains in the same carrier fiber and cannot weaken the lower bound.

Corollary 18.2 (Weinberg benchmark is internally live). *q_W is a live complete internal realization residue.*

19 Explicit complete assignment for an active three-singlet benchmark

19.1 A concrete full-rank active point

Choose a complex symmetric full-rank low-energy coefficient

$$\kappa_* = UDU^T$$

with $D = \text{diag}(d_1, d_2, d_3)$, $d_i > 0$, nontrivial mixing U , and response-mode separations satisfying the frozen target. Choose any positive nonsingular diagonal heavy matrix

$$M_R = \text{diag}(M_1, M_2, M_3)$$

and set

$$Y_\nu = UD^{1/2}M_R^{1/2}. \quad (34)$$

Then

$$Y_\nu M_R^{-1} Y_\nu^T = \kappa_*. \quad (35)$$

Both Y_ν and M_R are full rank, every column of Y_ν is nonzero, and all three retained singlets are source active. Gauge singlets contribute zero to the Standard Model gauge-anomaly ledger. Together with the PCR–4 common vacuum, channel, response, matching, and failure packet, this gives a concrete point x_{3N} in the active local/hidden survivor locus.

19.2 No inactive padding and no assumed heavy-sector cell count

The parent trace of x_{3N} contains three active primitive singlet nodes, nonzero attachment incidence, nonsingular Majorana mass data, and the full heavy-to-visible continuation. Because every singlet participates

in (34), none is spectator padding.

Choose any

$$N_{3N} \in \mathcal{N}_5(x_{3N}).$$

Its role-bearing core contains the complete source-to-visible mandatory witness paths. The one-block factorization proves $\text{Fact}_5(N_{3N}) \neq \emptyset$, and finite maximality supplies

$$F_{3N} \in \text{MaxFact}_5(N_{3N}).$$

The construction does not infer that the heavy sector is one block merely because no subblock decouples. If basis-covariant spectral or Takagi subspaces satisfy all factorization clauses, they appear as a finer admissible factorization and are retained. Named weak-basis singlets alone do not define invariant blocks.

Theorem 19.1 (Three-singlet benchmark complete assignment). *Let $P_{3N} = P_{N_{3N}, F_{3N}}$, let $A_{3N} = A_{N_{3N}, F_{3N}}$, and let*

$$\sigma_{3N} : A_{3N} \xrightarrow{\sim} \Omega_5(N_{3N}, F_{3N})$$

be the full-factor address bijection. Then

$$z_{3N} = (x_{3N}, N_{3N}, F_{3N}, P_{3N}, \sigma_{3N}, e_{3N}, j_{3N}) \in \Sigma_5^{\text{int}}.$$

Proof. Every factor block has nonempty role-standing signature, is active at the fixed point, and belongs to a mandatory source-to-interface path; hence every E_5 clause passes. The complete factor graph reconstructs the same full-rank source-to-visible witness, uses one parent and parameter point, retains all constitutive maps, and inherits one PCR-4 common cone and one consistency/failure ledger; hence $J_5 = 1$. The assignment covers the entire maximal factorization. $\square$

Define

$$q_{3N} = [z_{3N}]_{\Gamma_5^{\text{int}}} \in \mathfrak{D}_5^{\text{int}}.$$

Its parent trace contains three active primitive singlet nodes and nonzero local attachment incidence. Its exact maximal-factorization block count and pattern are calculated data, not prerequisites of the endpoint.

Corollary 19.2 (Three-singlet benchmark is internally live). *q_{3N} is a live complete internal realization residue.*

20 Why low-energy matching does not collapse the benchmarks

20.1 Typed matching arrow

The explicit points in sections 18 and 19 can share the same low-energy coefficient:

$$\kappa_W = Y_\nu M_R^{-1} Y_\nu^T = \kappa_*. \quad (36)$$

This is a lawful EFT relation between overlap observables. It is not an isomorphism of complete internal assignment invariants.

Theorem 20.1 (No matching-induced internal morphism). *Even when (36) holds exactly,*

$$\text{Mor}_{\Gamma_5^{\text{int}}}(z_W, z_{3N}) = \emptyset.$$

Proof. The parent traces differ. x_W has no added primitive local neutral-fermion node; x_{3N} has three active singlet nodes, nonzero local attachment incidence, and heavy-sector continuation/failure data. A Γ_5^{int} -morphism must preserve the complete carrier trace. The exact factorization patterns need not be

compared: the carrier invariant already forbids a morphism. Equation (36) preserves only a coarser low-energy projection. $\square$

Corollary 20.2 (Benchmark orbit separation).

$$q_W \neq q_{3N}.$$

20.2 Matching has direction

The heavy-to-light arrow forgets primitive degrees of freedom, source resolution, ultraviolet continuation, and failure conditions. There is no inverse determined by κ_* ; many (Y_ν, M_R) pairs produce the same coefficient. Treating matching as identity would erase the parent carrier trace and manufacture a nonexistent inverse.

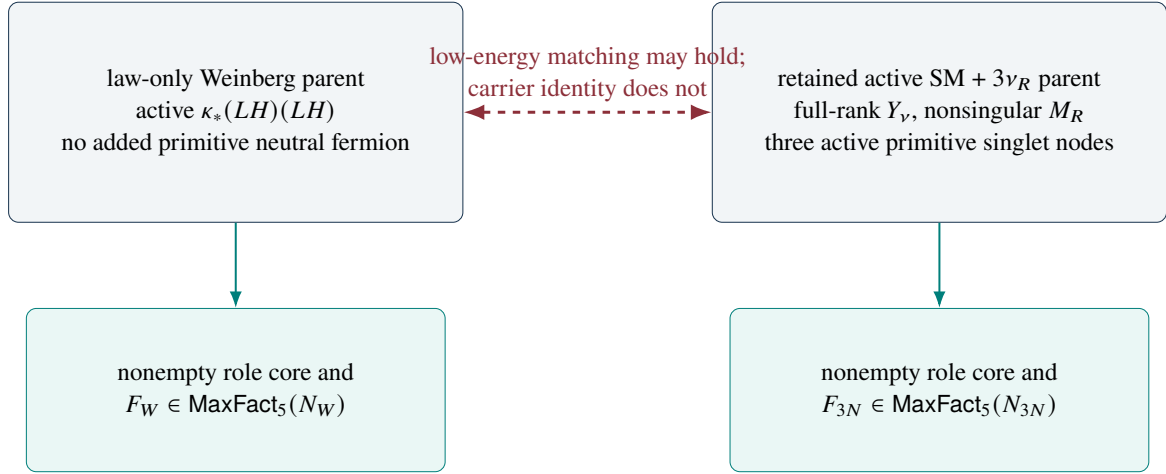


Figure 6: The explicit complete assignments may share a low-energy coefficient while retaining different parent carrier traces. Their exact maximal factorizations are not needed for separation.

21 Carrier-indexed internal orbit fibers and carrier-semantic descent

Because $\bar{\pi}_{\text{car}} : \mathfrak{D}_5^{\text{int}} \rightarrow \mathfrak{S}_\nu^{(4)}$ is well defined, set

$$\mathfrak{F}_5([x]) = \bar{\pi}_{\text{car}}^{-1}([x]). \quad (37)$$

This is the set of factorization-sensitive internal occupancy orbits over one frozen PCR-4 carrier class.

Lemma 21.1 (Every carrier fiber is nonempty). *For every $[x] \in \mathfrak{S}_\nu^{(4)}$, $\mathfrak{F}_5([x]) \neq \emptyset$.*

Proof. Choose a representative x . By theorem 13.2, $\Sigma_5^{\text{int}}(x) \neq \emptyset$. The orbit of any complete assignment over x lies in the displayed fiber. $\square$

Theorem 21.2 (Carrier-indexed fiber decomposition).

$$\mathfrak{D}_5^{\text{int}} = \bigsqcup_{[x] \in \mathfrak{S}_\nu^{(4)}} \mathfrak{F}_5([x]), \quad \mathfrak{F}_5([x]) \neq \emptyset. \quad (38)$$

Proof. Every internal orbit has one parent carrier class, so it lies in one fiber. Distinct fibers are disjoint because every internal groupoid morphism preserves the complete parent trace. Nonemptiness is lemma 21.1. $\square$

Corollary 21.3 (Universal carrier lower bound). *In the declared ZFC foundation,*

$$|\mathfrak{D}_5^{\text{int}}| \geq |\mathfrak{S}_\nu^{(4)}| \geq 2. \quad (39)$$

Proof. The carrier projection is a surjection from $\mathfrak{D}_5^{\text{int}}$ onto $\mathfrak{S}_\nu^{(4)}$. The first inequality is the standard cardinal comparison for a surjection. PCR-4 supplies at least two carrier-inequivalent survivor classes. $\square$

Theorem 21.4 (Universal occupancy lifting at carrier resolution). *Define the native carrier-semantic PCR-5 residue by*

$$\mathfrak{S}_{\nu, \text{car}}^{(5)} := \text{im}(\bar{\pi}_{\text{car}}). \quad (40)$$

Then

$$\boxed{\mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_\nu^{(4)}, \quad |\mathfrak{S}_{\nu, \text{car}}^{(5)}| \geq 2.} \quad (41)$$

Thus compatible occupancy universally lifts every PCR-4 carrier survivor; it does not eliminate a carrier class at the internal stage.

Proof. The image is contained in $\mathfrak{S}_\nu^{(4)}$ by construction. Every $[x] \in \mathfrak{S}_\nu^{(4)}$ has a nonempty fiber by lemma 21.1, so every such class occurs in the image. The cardinal lower bound is inherited from PCR-4. $\square$

Remark 21.5 (No canonical section and no factorization promotion). *No preferred map $\mathfrak{S}_\nu^{(4)} \rightarrow \mathfrak{D}_5^{\text{int}}$ is defined. A carrier fiber may contain several normal-form or maximal-factorization orbits. Those differences are retained by the internal classifier but are not promoted into separate terminal carrier residues merely because their factorization graphs differ.*

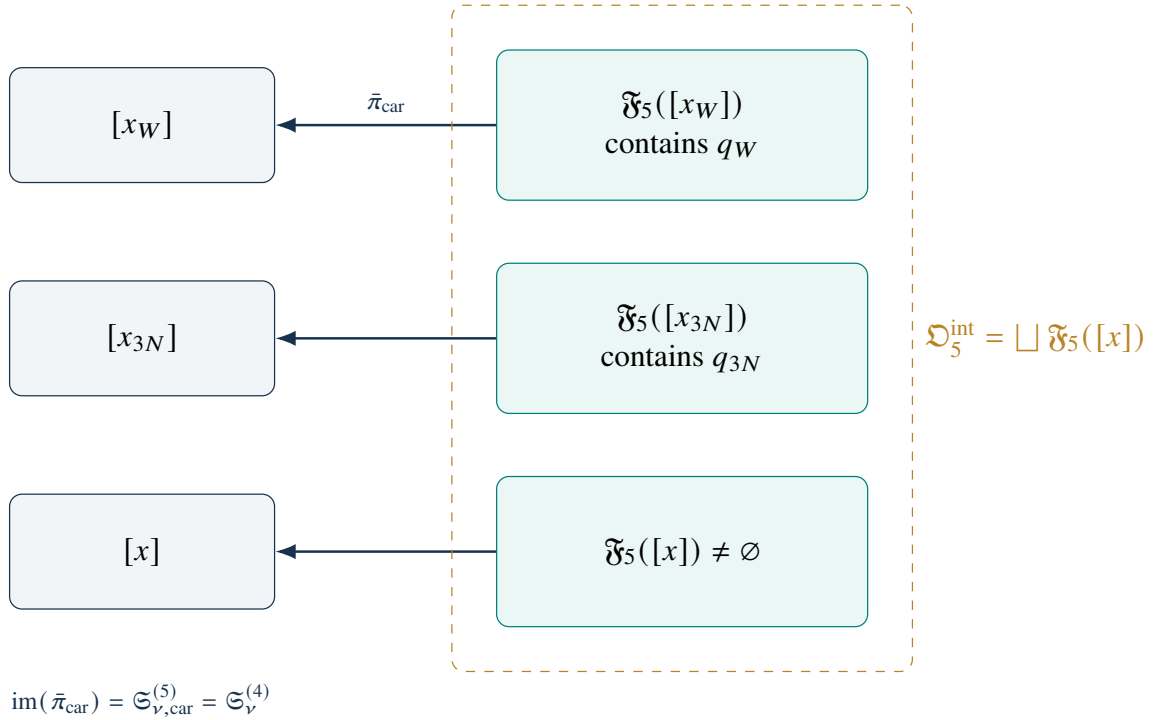


Figure 7: Every PCR-4 carrier class indexes a nonempty internal occupancy fiber. The terminal carrier-resolution residue is the carrier projection of these fibers, not their factorization-sensitive orbit count.

Proposition 21.6 (Additional factorizations cannot restore carrier singletonhood). *Additional valid normal forms, maximal factorizations, or internal assignments can enlarge an internal fiber but cannot merge two frozen carrier-semantic classes. In particular they cannot identify the law-only and active-three-singlet parents while χ_{car} remains invariant.*

22 Internal endpoint calculation

22.1 Branch structure

The internal classification has three possible branches:

$$|\mathfrak{D}_5^{\text{int}}| = 0, \quad |\mathfrak{D}_5^{\text{int}}| = 1, \quad |\mathfrak{D}_5^{\text{int}}| > 1. \quad (42)$$

They mean respectively no complete role-bearing factorization assignment, one factorization-sensitive internal occupancy class, and multiple factorization-sensitive occupancy classes. The carrier-semantic endpoint is computed separately through the carrier projection.

The explicit assignments satisfy

$$z_W, z_{3N} \in \Sigma_5^{\text{int}}, \quad q_W = [z_W]_{\Gamma_5^{\text{int}}} \neq [z_{3N}]_{\Gamma_5^{\text{int}}} = q_{3N}. \quad (43)$$

Theorem 22.1 (Strict factorization-sensitive internal orbit multiplicity).

$$|\mathfrak{D}_5^{\text{int}}| \geq |\mathfrak{S}_\nu^{(4)}| \geq 2, \quad |\mathfrak{D}_5^{\text{int}}| > 1. \quad (44)$$

Proof. The universal inequality is corollary 21.3. Independently, theorems 18.1 and 19.1 and corollary 20.2 give two explicit distinct orbit elements. Neither argument assumes exact benchmark factor counts. $\square$

Corollary 22.2 (Zero and singleton branches are excluded).

$$|\mathfrak{D}_5^{\text{int}}| \neq 0, \quad |\mathfrak{D}_5^{\text{int}}| \neq 1.$$

Table 3: Internal endpoint branches.

Branch	Required condition	Disposition
Zero	No complete maximal-factorization assignment.	Excluded by every nonempty carrier fiber.
Singleton	All complete assignments lie in one carrier-preserving orbit.	Excluded by $q_W \neq q_{3N}$.
Multiple	At least two lawful internal orbits survive.	Realized.

22.2 Lower bound and exact cardinality

The theorem does not prove $|\mathfrak{D}_5^{\text{int}}| = 2$. Composite, solitonic, environmental, boundary, hybrid, parameter, normal-form, or factorization fibers may add orbits. The strongest internal statement is strict factorization-sensitive multiplicity with the universal carrier lower bound. Terminal carrier multiplicity is the separate projection result $\mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_\nu^{(4)}$.

Theorem 22.3 (Endpoint stability under additional factorizations). *Adding valid maximal factorizations or assignments while preserving the groupoid invariant cannot change the branch from multiple to singleton.*

Proof. The two benchmark orbits remain present and cannot be merged by a carrier-preserving morphism. New objects can only preserve or increase the orbit cardinality. $\square$

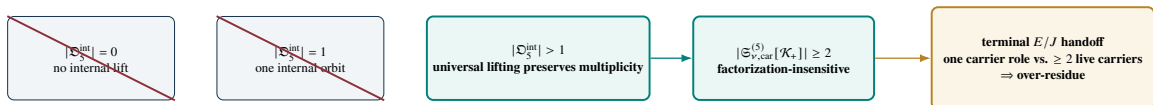


Figure 8: The internal factorization-sensitive classifier is multiple. Terminal ROC is applied only after carrier-semantic descent and explicit terminal eligibility/compatibility formation.

23 Cross-corpus selector analysis and stability

23.1 Bounded reference corpus

The bounded reference corpus comprises the fixed PCR–1–PCR–4 sequence, the Standard Model carrier/cardinality arc, the SYM reference arc, the neutrino support corpus, the Standard-Model Realization Rigidity arc, the Exactness arc, and the additional proof-level sources used below. The supplementary source register records eighty PDFs spanning 5,099 pages, with seventy-nine unique document payloads. The generation import used here is the primitive-standing reference edition identified below; the register totals describe the recorded corpus inventory.

The relevant question is whether any result already established in this corpus defines a lawful predicate on the unchanged PCR–5 target that either removes one benchmark assignment or identifies the two parent carrier traces.

Definition 23.1 (Lawful current-target selector). *A predicate D from $\mathcal{K}_+$ is a lawful current-target selector when:*

1. *it is defined on $(\mathfrak{C}_{\nu,\text{int}}^{(5)}, T_{\nu,\text{int}}^{(5)})$, not on a substituted minimal-SM, low-energy-only, background-probe, or enlarged-observable target;*
2. *its discriminating coordinate has independent standing before benchmark inspection;*
3. *it is total on both q_W and q_{3N} at the same retained resolution;*
4. *it changes admissibility rather than merely labeling different target fibers;*
5. *it is invariant under Γ_5^{int} and preserves parent-carrier resolution;*
6. *it respects dependency order.*

23.2 Corpus-aligned three-singlet point

PCR–4 proves a nonempty active local/hidden locus for admitted singlet counts $n \geq 2$, so the full-rank specialization $n = 3$ used in section 19 is lawful [8]. Write

$$q_{3N} = [\mathfrak{z}_{3N}]_{\Gamma_5^{\text{int}}}. \quad (45)$$

The SYM atlas places it in

$$D_{\text{NURM}} = \{\text{SM} + 3\nu_R, M_R \neq 0, d \leq 4\}, \quad (46)$$

and keeps that fiber distinct from

$$D_{\text{NUW}} = \{\text{minimal SM without dynamical } \nu_R, \kappa \neq 0, d \leq 5\} \quad (47)$$

[13]. Heavy-Majorana matching is an inter-fiber EFT transport relation, not representative equivalence.

The primitive-standing generation theorem proves

$$\text{ClosedGenCert}_{\mathcal{P}_0}(n) \iff n = 3 \quad (48)$$

for complete occupation of the independently constructed generation roles [2, Theorem 5.4]. Its count-independent primitive necessity theorem is $\text{Prim}_{\mathcal{P}_0}(I) \iff |I| = 3$ [2, Theorem 6.2]; joint constructibility and the absence of a proper adequate support define that qualification. Rank one is a consequence of the derived architecture, not an imported count premise. Additional ultraviolet data supply no additional primitive generation role for unchanged $\mathcal{P}_0$ [2, Theorem 6.4].

Proposition 23.2 (Three-singlet specialization). *$q_{3N} \in \mathfrak{D}_5^{\text{int}}$ is represented by a complete maximal-factorization assignment and is carrier inequivalent to q_W .*

Proof. This is corollaries 19.2 and 20.2. The constructive full-rank source point has a nonempty role core and at least one maximal admissible factorization; the parent trace retains three active singlet nodes. No exact factor count is required. $\square$

23.3 Fixed-load Standard-Model results

The Standard-Model carrier papers close primitive matter and bookkeeping only inside their fixed same-scope target [3, 12]. PCR-1 deliberately leaves neutrino carrier type and completion mechanism open, PCR-3 generates those completions, and PCR-4 evaluates them. Backflowing a minimal-SM closure to delete an admitted active completion would replace the target rather than apply a same-target selector.

The Realized Physical Interior likewise records the neutrino mechanism as an open status coordinate [11]. The Structura and charge-lattice results force common charged-sector structure while admitting a gauge-singlet $Y = 0$ extension and leaving the neutrino mass mechanism downstream [6, 9]. The SR1–SR7 endpoint is a common mass-pre-dynamical Standard-Model-like gauge skeleton; it does not choose between the two benchmark source architectures.

23.4 Low-energy projection and constructive lift

Let

$$\pi_{\text{LE}} : \mathfrak{D}_5^{\text{int}} \longrightarrow Q_{\text{LE}} \quad (49)$$

forget heavy-node inventory while retaining the active light-neutrino propagation object and admitted observables. The exact-character theorem closes its own target to $\text{MajoranaCompatible}_{\text{LE}}$ while explicitly refusing to select ultraviolet origin [10]. The standing splitting ratio lies in

$$0.0299486814926010 \leq \rho_\nu \leq 0.0299486814926012, \quad (50)$$

but this numerical object also factors through π_{LE} [4, 14].

Lemma 23.3 (Three-singlet lift). *For every complex symmetric 3×3 low-energy coefficient κ_* , there exist nonsingular M_R and a Y_ν such that*

$$Y_\nu M_R^{-1} Y_\nu^T = \kappa_*. \quad (51)$$

The construction can be chosen full rank on the nondegenerate stratum.

Proof. Take a Takagi factorization $\kappa_* = UDU^T$, choose any positive nonsingular diagonal M_R , and set $Y_\nu = UD^{1/2}M_R^{1/2}$. Then $Y_\nu M_R^{-1} Y_\nu^T = UDU^T$. On a full-rank D , all factors can be chosen full rank. $\square$

Thus low-energy character, ordering, PMNS-invariant data, and the standing ratio can be common to both internal fibers.

23.5 Symmetry and $B - L$ disposition

The symmetry corpus distinguishes local polynomial anomaly behavior: one ν_R per family can cancel the cubic and mixed-gravitational $B - L$ deficits, whereas the minimal no- ν_R fiber does not [1]. This is a genuine potential discriminator coordinate, but the frozen PCR-5 target does not activate the complete background/global form, breaking, line/defect, and global-anomaly packet. Generic M_R and the Weinberg operator also break continuous $B - L$. Local cancellation is therefore a successor-target burden, not a current selector.

23.6 RG transport and Exactness

Standing-preserving RG and threshold matching preserve overlap observables but do not identify ultraviolet field inventories [5]. The EX1–EX9 arc forbids promotion of a singleton quotient, numerical residue, transport orbit, cross-sector aggregate, or extraction output into a singleton raw carrier origin while distinct non-skin parent representatives remain.

Corollary 23.4 (No carrier-origin promotion). *A singleton value of π_{LE} , a numerical readout, or an exact transported residue does not imply $q_W = q_{3N}$.*

Proof. The projection fiber contains at least the two carrier-distinct benchmark orbits. Exactness of the image does not collapse the nontrivial parent fiber; that is precisely the quotient-fiber obstruction formalized in the Exactness arc. $\square$

23.7 Selector classification

The complete twenty-nine-row selector classification is included in the supplementary data. The load-bearing classes are summarized below.

Table 4: Representative cross-corpus selector dispositions.

Rows	Imported result class	Disposition	Reason
C01–C05	PCR target, carrier, generation, and fixed-load Standard-Model governance	Common constraint or target guard	Applies equally or is scoped to a different fixed target.
C06–C09	SYM target fibers and low-energy exact character	Distinction or coarse projection	Distinguishes descriptions or common residue; does not select.
C10–C11, C16	Splitting ratio, ordering, and low-energy witnesses	Common numerical/observable projection	Constructive three-singlet lift realizes the same data.
C12–C13, C15	$B - L$, heavy-state, nonunitarity, flavor, collider, cosmological, dimension-six coordinates	Future target	Requires a rebuilt target and completeness theorem.
C14	Simplicity, field count, operator dimension, naturalness	Forbidden preference	No independent target standing.
C17	Additional family, normal-form, and factorization fibers	Upper-closure burden	Can add orbits; cannot delete existing witnesses.
C18–C19	Role-Occupancy Closure	Terminal diagnostic	Supplies the fixed-role closure theorem; terminal eligibility, compatibility, and live-residue formation are discharged separately in PCR–5.
C20–C24	Realized interior, fixed bookkeeping, charge lattice, RG matching, gauge rigidity	Common or transported	Does not identify carrier origin.
C25–C29	Exactness and extraction results	Anti-promotion theorem	Blocks inference from coarse exactness to raw carrier identity.

Theorem 23.5 (Corpus-disposition theorem). *No registered output in $\mathcal{K}_+$ satisfies all clauses of definition 23.1.*

Proof. The twenty-nine-row matrix assigns every selection-relevant output its source, target relation, values on the two benchmarks, and logical disposition. The preceding subsections prove the five exhaustive dispositions: common constraint, coarser projection, target-fiber distinction, future-target burden, or forbidden selector. None both preserves the current target and deletes or identifies one benchmark. $\square$

23.8 Restricted internal object

Let $\mathcal{K}_+^{\text{cur}}$ contain only predicates well typed on the unchanged internal target. Define

$$\Sigma_5^{\text{int}}[\mathcal{K}_+] = \{z \in \Sigma_5^{\text{int}} : C(z) = 1 \text{ for every } C \in \mathcal{K}_+^{\text{cur}}\}, \quad (52)$$

$$\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+] = \Sigma_5^{\text{int}}[\mathcal{K}_+] / \left(\Gamma_5^{\text{int}}|_{\Sigma_5^{\text{int}}[\mathcal{K}_+]} \right). \quad (53)$$

Theorem 23.6 (Cross-corpus stability of strict multiplicity).

$$q_W, q_{3N} \in \mathfrak{D}_5^{\text{int}}[\mathcal{K}_+], \quad q_W \neq q_{3N}, \quad \boxed{|\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]| \geq 2}. \quad (54)$$

Proof. Both explicit assignments pass the common same-target constraints. The remaining imports are coarser projections, distinct target fibers, future-target coordinates, or forbidden preferences by theorem 23.5. The restricted groupoid still preserves parent carrier trace and role-core/maximal-factorization architecture, so the two orbits remain distinct. $\square$

Corollary 23.7 (Bounded carrier-semantic residue remains non-singleton). *Define*

$$\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+] := \bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]). \quad (55)$$

Then

$$\boxed{|\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2}. \quad (56)$$

Proof. The two surviving benchmark orbits lie in $\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]$ and have distinct parent carrier traces. Their images under $\bar{\pi}_{\text{car}}$ are therefore two distinct carrier-semantic classes in $\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$. Possible additional factorization-sensitive orbits over either parent do not increase the number of terminal carrier residues unless the parent carrier class changes. $\square$

Corollary 23.8 (No current-corpus singleton selector). *The bounded reference corpus does not turn either the restricted internal occupancy object or its carrier-semantic projection $\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$ into a singleton.*

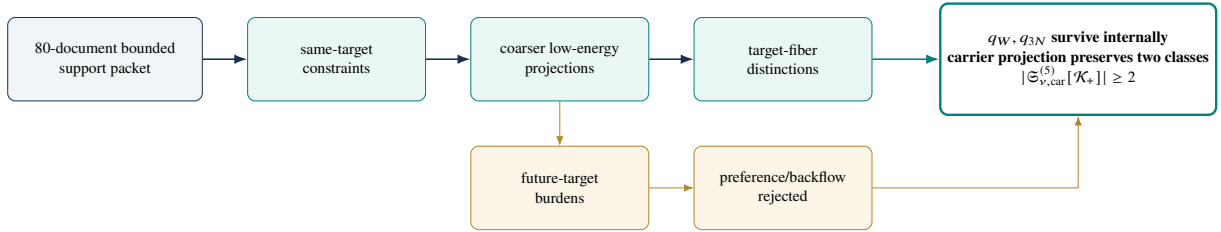


Figure 9: The bounded reference corpus preserves both explicit internal orbit fibers.

23.9 Scope closure of the bounded analysis

Every selection-relevant result in the declared reference corpus has a target-relative disposition. The bounded analysis therefore establishes

$$\text{all applicable corpus constraints} \not\Rightarrow \text{unique carrier origin}. \quad (57)$$

This does not assert unrestricted future semantic exhaustion. Direct heavy-state, correlated ultraviolet, or complete global $B - L$ information would define a distinct successor target.

23.10 Primitive generation receipt at retained carrier resolution

Proposition 23.9 (Generation receipt preserves the two carrier targets). *Receive a common primitive generation datum on the Standard Model input of the two benchmark constructions, with a joint injective assignment covering its primitive live occurrence set. Both benchmarks then have three primitive generation positions. This receipt neither fixes their added-singlet count nor identifies their complete carrier traces or their maximal-factorization inventories.*

Proof. The generation datum receives the inherited matter/charge/weak/bridge licenses, complete comparison modulo rephasing, and a joint capability witness. A positive common datum is supplied by the charged sector of [2, Proposition 4.4], together with its admitted boundary-capability family. Its intrinsic occurrence projections, pair readers and loop readers retain the original primitive meanings; the source’s Theorems 6.2 and 5.4 give three positions and complete occupation. The admission of this common datum is part of the inherited Standard Model source input, not a consequence of the dimension of Y_ν or M_R .

The law-only construction retains these Standard Model source and boundary incidences and supplies the actual admitted coefficient κ_* . The active-singlet construction retains the same charged-sector incidence and adds source-active singlets with $Y_\nu M_R^{-1} Y_\nu^T = \kappa_*$. The additional singlets are neutral under the Standard Model gauge group, so they do not alter its charged-sector anomaly equations or the received primitive assignment. PCR-4 supplies the separate joint gauge, vacuum, continuation and common-cone receipt used in both internal-occupancy proofs. The actual propagation parameters and their outcomes remain those of each benchmark; the capability receipt is not substituted for those outputs.

Thus the generation-count reader takes the same value on both receipts. The primitive datum does not contain the parent-trace reader for added singlet nodes: that reader takes zero on the law-only parent and three on the active-singlet parent. It cannot factor through a receipt which identifies their generation data. Their carrier inequivalence from corollary 20.2 therefore remains intact. The factor-block inventory and terminal complete-carrier role are separately typed objects; no map identifying them with generation positions is supplied or needed. $\square$

The benchmark’s three-singlet choice is licensed by the PCR-4 local construction and the SYM target fibre. It is not derived from generation cardinality. The source’s shared charged-multiplicity theorem leaves hypercharge-zero singlet multiplicity unconstrained [2, Proposition 4.1]. A proposed interpretation of any further original reader must preserve its actual domain and be constant on the claimed receipt fibres [2, Proposition 6.7]; agreement of the generation count alone gives no ultraviolet carrier equivalence.

24 Terminal Role-Occupancy Closure and the over-residue theorem

24.1 Terminal carrier resolution

The imported Role-Occupancy Closure theorem fixes a role set before residue matching, forms a same-certificate eligibility relation, tests full-assignment compatibility, defines the live residue as the union of images of injective eligible compatible preassignments, and only then asks for a compatible bijection onto that live residue [7]. PCR-5 follows this order at carrier-semantic resolution.

The internal object $\mathfrak{D}_5^{\text{int}}$ is factorization-sensitive. The terminal target is instead carrier-semantic, so its native candidate residue is the frozen PCR-4 carrier quotient

$$\text{Residue}_{\text{term}} := \mathfrak{S}_\nu^{(4)}. \quad (58)$$

By theorems 16.4 and 21.4, all factorization-sensitive variants over one parent carrier descend to the same terminal carrier residue, while carrier-inequivalent parents remain distinct.

The terminal target and its one-role architecture were declared in section 3 before the internal count. By theorem 3.2,

$$R_5^{\text{term}} = \{r_{\text{car}}(T_\nu^{\text{osc}})\}, \quad |R_5^{\text{term}}| = 1.$$

24.2 Terminal eligibility relation

Define the terminal role-eligibility relation

$$E_5^{\text{term}} \subseteq \text{Residue}_{\text{term}} \times R_5^{\text{term}} \quad (59)$$

by

$$E_5^{\text{term}}([x], r_{\text{car}}(T_{\nu}^{\text{osc}})) = 1 \iff [x] \in \mathfrak{S}_{\nu}^{(4)}. \quad (60)$$

Thus

$$\text{Residue}_{\text{term}}^{\text{elig}} = \mathfrak{S}_{\nu}^{(4)}. \quad (61)$$

Proposition 24.1 (Formation of terminal eligibility edges). *Every positive edge in (60) is generated by pre-existing target architecture and carrier provenance, not by the endpoint count.*

Proof. An element of $\mathfrak{S}_{\nu}^{(4)}$ is, by the frozen PCR-3/PCR-4 construction, a presentation-quotiented same-target carrier survivor for T_{ν}^{osc} . The terminal role asks precisely for a carrier-semantic realization of that same target. Hence the eligibility edge is internal role typing plus provenance/indexical typing in the sense of the ROC formation rules. No likelihood, simplicity ranking, empirical post-selection, or factorization count is used. $\square$

24.3 Terminal compatibility predicate

For a native terminal singleton assignment candidate

$$\tau_{[x]} : R_5^{\text{term}} \hookrightarrow \text{Residue}_{\text{term}}^{\text{elig}}, \quad \tau_{[x]}(r_{\text{car}}(T_{\nu}^{\text{osc}})) = [x],$$

define

$$J_5^{\text{term}}(\tau_{[x]}) = 1 \iff \mathfrak{F}_5([x]) \neq \emptyset. \quad (62)$$

For the bounded same-target update, define

$$J_5^{\text{term}, \mathcal{K}_+}(\tau_{[x]}) = 1 \iff \{q \in \mathfrak{D}_5^{\text{int}}[\mathcal{K}_+] : \bar{\pi}_{\text{car}}(q) = [x]\} \neq \emptyset. \quad (63)$$

These predicates state that the proposed carrier residue possesses at least one complete internally eligible and globally compatible role-occupancy realization at the relevant certificate level.

Theorem 24.2 (Internal-to-terminal eligibility/compatibility handoff). *For every $[x] \in \mathfrak{S}_{\nu}^{(4)}$,*

$$E_5^{\text{term}}([x], r_{\text{car}}(T_{\nu}^{\text{osc}})) = 1, \quad J_5^{\text{term}}(\tau_{[x]}) = 1. \quad (64)$$

Moreover, for every $[x] \in \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$,

$$E_5^{\text{term}}([x], r_{\text{car}}(T_{\nu}^{\text{osc}})) = 1, \quad J_5^{\text{term}, \mathcal{K}_+}(\tau_{[x]}) = 1. \quad (65)$$

Proof. Terminal eligibility is proposition 24.1. Native compatibility follows from lemma 21.1: every PCR-4 carrier survivor has at least one complete internal E_5/J_5 -positive assignment. For the bounded statement, membership in $\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+] = \bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+])$ is exactly the existence of at least one bounded-support internal orbit above $[x]$, so (63) holds. The formation source for J_5^{term} is the certificate-declared internal co-instantiation theorem: the factorized assignment already preserves the common cone, continuation, boundary, failure, and one-survivor conditions. Thus internal E_5/J_5 is not silently substituted for terminal E/J ; it supplies the proved witness used by the separately defined terminal compatibility predicate. $\square$

24.4 Preassignments and exact live residue

The native and bounded terminal preassignment spaces are

$$\Sigma_5^{\text{pre,term}} = \{\tau_{[x]} : [x] \in \mathfrak{S}_\nu^{(4)}, E_5^{\text{term}}([x], r_{\text{car}}(T_\nu^{\text{osc}})) = 1, J_5^{\text{term}}(\tau_{[x]}) = 1\}, \quad (66)$$

$$\Sigma_5^{\text{pre,term}}[\mathcal{K}_+] = \{\tau_{[x]} : [x] \in \mathfrak{S}_\nu^{(4)}, E_5^{\text{term}}([x], r_{\text{car}}(T_\nu^{\text{osc}})) = 1, J_5^{\text{term}, \mathcal{K}_+}(\tau_{[x]}) = 1\}. \quad (67)$$

The ROC live residues are therefore

$$\text{Residue}_{\text{term}}^{\text{live}} := \bigcup_{\tau \in \Sigma_5^{\text{pre,term}}} \text{im } \tau = \mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_\nu^{(4)}, \quad (68)$$

$$\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] := \bigcup_{\tau \in \Sigma_5^{\text{pre,term}}[\mathcal{K}_+]} \text{im } \tau = \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]. \quad (69)$$

Theorem 24.3 (Exact terminal live-residue calculation). *Equations (68) and (69) are the live residues obtained from the ROC definitions of eligibility, compatibility, and preassignment. Factorization-sensitive internal orbits over one carrier do not appear as multiple terminal residues.*

Proof. By theorem 24.2, the native preassignments are exactly the singleton maps to the carrier classes having nonempty native internal fibers; theorem 21.4 says this set is all of $\mathfrak{S}_\nu^{(4)}$. The bounded preassignments are exactly those whose parent carrier occurs in the image of $\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]$, namely $\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$. Taking the union of images gives the displayed sets. The carrier projection performs the terminal factorization-insensitive quotient before live-residue formation. $\square$

24.5 Terminal certificate completeness

Theorem 24.4 (Carrier-resolution terminal certificate completeness). *The native terminal certificate is role-realization complete for $T_{\nu, \text{term}}^{(5)}$ at the declared carrier resolution. The bounded $\mathcal{K}_+$ update is complete relative to its explicitly defined current-target restriction $\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]$.*

Proof. The seven ROC completeness burdens are discharged as follows.

1. *Candidate completeness:* PCR–3/PCR–4 supply the exact carrier-semantic survivor quotient $\mathfrak{S}_\nu^{(4)}$; the bounded update uses the explicitly defined carrier image of the restricted internal object.
2. *Quotient completeness:* the terminal residue uses the frozen carrier equivalence, while theorem 16.4 prevents maximal-factorization variants from being overcounted as distinct carrier residues.
3. *Role-set completeness:* R_5^{term} is a field of the separately declared terminal target and was fixed before residue inspection.
4. *Eligibility completeness:* E_5^{term} records exactly the same-target carrier-provenance edge generated by the terminal carrier role; there is no ranking-generated edge.
5. *Compatibility completeness:* J_5^{term} and $J_5^{\text{term}, \mathcal{K}_+}$ are formed from existence of complete internal E_5/J_5 -positive co-instantiation witnesses, whose compatibility grammar includes common-cone, continuation, boundary, failure, and ownership data.
6. *Live-residue completeness:* theorem 24.3 forms the live residue from all and only positive terminal preassignments. An eligible carrier with empty bounded fiber has the explicit diagnosis $J_5^{\text{term}, \mathcal{K}_+} = 0$, rather than remaining silent residue.
7. *Failure completeness:* PCR–4 exclusions and the bounded corpus dispositions fail closed at their declared scopes; neither certificate converts an unclassified preference into an occupancy edge.

Thus the ROC biconditional is applicable at the declared terminal resolution. $\square$

Terminal lawful automorphisms. At carrier resolution define the terminal assignment invariant

$$I_5^{\text{term}} = (\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)}, R_5^{\text{term}}, \text{Residue}_{\text{term}}^{\text{live}}, E_5^{\text{term}}, J_5^{\text{term}}, \chi_{\text{car}}, \mathcal{B}_{\text{term}}, \mathcal{C}_{\text{term}}, \mathcal{F}_{\text{term}}),$$

where the last three entries collect the terminal boundary, continuation, and failure data inherited from the certificate. The certificate-lawful terminal automorphism group is

$$\Gamma_5^{\text{term}} := \text{Aut}(I_5^{\text{term}}),$$

and the bounded certificate uses the corresponding restriction $\Gamma_5^{\text{term}}[\mathcal{K}_+]$. These automorphisms preserve the target, forced role, live residue, eligibility, compatibility, carrier-semantic standing, boundary, continuation, and failure structure. They therefore remove only lawful relabeling at the terminal role-incidence level.

24.6 Over-residue and closure

Because $\mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_{\nu}^{(4)}$ and $|\mathfrak{S}_{\nu}^{(4)}| \geq 2$, while corollary 23.7 gives $|\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2$,

$$|\text{Residue}_{\text{term}}^{\text{live}}| \geq 2 > 1 = |R_5^{\text{term}}|, \quad |\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+]| \geq 2 > 1. \quad (70)$$

Theorem 24.5 (Terminal role-covering failure). *For both the native and bounded-support certificates,*

$$\Sigma_5^{\text{term}} = \emptyset, \quad \Sigma_5^{\text{term}}[\mathcal{K}_+] = \emptyset, \quad (71)$$

where Σ_5^{term} denotes the ROC set of compatible bijective role-covering assignments from R_5^{term} onto the complete live residue.

Proof. Every terminal role-covering assignment must be a bijection from the singleton set R_5^{term} onto the corresponding live residue. Equation (70) rules out such a bijection. The compatible preassignments are nonempty, but the terminal role-covering assignment space is empty because more than one live carrier-semantic alternative remains. $\square$

Consequently, for the lawful terminal actions just defined,

$$\Sigma_5^{\text{term}} / \Gamma_5^{\text{term}} = \emptyset, \quad \Sigma_5^{\text{term}}[\mathcal{K}_+] / \Gamma_5^{\text{term}}[\mathcal{K}_+] = \emptyset.$$

Theorem 24.6 (PCR–5 compatible role-occupancy capstone). *For the frozen PCR imports and bounded support packet,*

$$\boxed{\mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_{\nu}^{(4)}, \quad |\mathfrak{S}_{\nu, \text{car}}^{(5)}| \geq 2, \quad |\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2, \quad \neg \text{RealizationLicensed}(\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)})}. \quad (72)$$

The terminal diagnosis is over-residue: compatible carrier realizations exist, but the separately declared one-role carrier-resolution target does not close to a unique licensed residue.

Proof. Universal occupancy lifting is theorem 21.4; bounded carrier multiplicity is corollary 23.7. Terminal role, eligibility, compatibility, and live-residue formation are established by theorems 3.2, 24.2 and 24.3. The terminal certificate is role-realization complete by theorem 24.4. By theorem 24.5, the compatible role-covering assignment set is empty, so its lawful-orbit quotient is not singleton. The Role-Occupancy Closure biconditional therefore yields $\neg \text{RealizationLicensed}(\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)})$. $\square$

24.7 Role-Occupancy Closure conditions

Table 5: Role-Occupancy Closure conditions.

Clause	Burden	Discharge
C1	Frozen upstream target and carrier residue.	PCR–1–PCR–4 and section 2.
C2	Internal role-bearing occupancy lift.	Sections 5, 6 and 13.
C3	Factorization-sensitive internal quotient.	Sections 15 and 16.
C4	Factorization-insensitive carrier-resolution descent.	Theorems 16.4 and 21.4.
C5	Terminal role set fixed before residue inspection.	Theorem 3.2.
C6	Terminal eligibility relation E_5^{term} .	Proposition 24.1.
C7	Terminal compatibility J_5^{term} and handoff.	Theorem 24.2.
C8	Preassignment space and live-residue union.	Theorem 24.3.
C9	Terminal certificate completeness.	Theorem 24.4.
C10	Bounded carrier-semantic multiplicity.	Corollary 23.7.
C11	Compatible role-covering bijection.	Fails by theorem 24.5; this is the calculated over-residue branch.

Corollary 24.7 (No current-corpus unique license). *No supplied same-target theorem makes $\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$ singleton or supplies a lawful cross-carrier equivalence. Hence the bounded reference corpus does not remove the terminal over-residue.*

Closure consequence

PCR–5 establishes universal compatible occupancy lifting, a carrier-semantic residue with at least two classes, and terminal ROC over-residue. Factorization alternatives over one parent are not counted as distinct terminal carrier residues. The failure of unique realization licensing therefore survives the strongest factorization-insensitive carrier-resolution quotient supported by the frozen theorem chain.

25 Structural clarifications and limiting cases

25.1 Common-cone coherence and factor count

A common cone proves co-instantiation and transfers J_5 to the complete factorized diagram. It does not determine whether a maximal admissible factorization has one block, several blocks, or several incomparable maximal factorizations.

25.2 Why common refinement is not assumed

Two admissible factorizations may be supported by role-standing properties of different composite blocks while their common refinement contains subblocks with no independent role standing. The analysis therefore

retains every maximal element of the finite admissible-refinement poset rather than replacing incomparable factorizations by an unproved common refinement.

25.3 Maximality and uniqueness

“Maximal” means that no proper admissible refinement exists; it does not mean greatest, canonical, or preferred. A normal form may have several incomparable maximal factorizations, each defining an internal factorization-sensitive assignment.

25.4 Internal factorization and terminal carrier identity

Factorization belongs to the internal occupancy problem, whereas the terminal target resolves only carrier-semantic identity. The terminal equivalence relation is the kernel of $\bar{\pi}_{\text{car}}$, so incomparable factorizations over one fixed carrier class descend to one carrier-semantic residue. Factorization structure therefore cannot by itself generate terminal carrier multiplicity.

25.5 Role-bearing core and auxiliary trace

Carrier identity, parameter, continuation, or failure information may be indispensable to the complete realization trace without constituting an occupancy block. Such information remains in $\text{Aux}_5(N)$ unless it lies on a mandatory role-standing path.

25.6 Role standing of factor blocks

Every admissible factor block has nonempty RSig_5 . Generic trace relevance is therefore not used as a surrogate for role incidence.

25.7 Benchmark factorization neutrality

The endpoint does not require the Weinberg benchmark to be one-block or the three-singlet benchmark to have exactly two blocks. Each benchmark requires only one complete maximal-factorization assignment. Carrier-trace inequivalence, not factor count, separates their terminal carrier residues.

25.8 Basis-covariant heavy-sector refinements

The three-singlet realization does not identify weak-basis labels with physical factor blocks and does not infer one heavy block from nondecoupling. Any basis-covariant spectral or Takagi subspace satisfying the factorization clauses appears as a legitimate finer factorization. Such refinement may change the internal classifier without changing the parent carrier residue.

25.9 Terminal eligibility and compatibility

Internal E_5 acts on factor blocks, whereas terminal E_5^{term} acts on carrier-semantic residues and is formed from same-target carrier provenance. Terminal J_5^{term} is separately required: it is positive only when the carrier class has at least one complete internal E_5/J_5 -positive realization at the relevant certificate level. This preserves the ROC distinction between pointwise eligibility and joint compatibility.

25.10 Terminal role architecture

The paper does not infer one role from one target instance. The carrier-resolution target is defined with one complete-carrier slot, and that role set is fixed before residue inspection.

25.11 Low-energy equality

A common κ , mass ordering, splitting ratio, or interface profile is a coarse projection. It may establish empirical equivalence at one resolution, but it does not supply an inverse to the heavy-to-light descent or erase the parent carrier trace.

25.12 Interpretation of terminal over-residue

The terminal candidate residue is the fixed carrier-semantic survivor quotient, not the factorization-sensitive internal orbit count. After explicit terminal eligibility and compatibility are evaluated, at least the law-only and active-three-singlet carrier classes remain live. The over-residue therefore survives after factorization overcount has been removed.

26 Falsifiability and scope of revision

The load-bearing dependency chain is

$$\begin{aligned}
 x \in \mathfrak{S}_v^{(4)} &\implies \mathcal{N}_5(x) \neq \emptyset \implies \text{Core}_5(N) \neq \emptyset \implies \text{MaxFact}_5(N) \neq \emptyset \\
 &\implies \Sigma_5^{\text{int}}(x) \neq \emptyset \implies \mathfrak{F}_5([x]) \neq \emptyset \implies \mathfrak{S}_{v,\text{car}}^{(5)} = \mathfrak{S}_v^{(4)}, \\
 q_W, q_{3N} \in \mathfrak{D}_5^{\text{int}}[\mathcal{K}_+], \quad \bar{\pi}_{\text{car}}(q_W) \neq \bar{\pi}_{\text{car}}(q_{3N}) &\implies |\mathfrak{S}_{v,\text{car}}^{(5)}[\mathcal{K}_+]| \geq 2, \\
 E_5^{\text{term}} + J_5^{\text{term}} &\implies \text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] = \mathfrak{S}_{v,\text{car}}^{(5)}[\mathcal{K}_+], \\
 |\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+]| \geq 2 > |\mathcal{R}_5^{\text{term}}| = 1 &\implies \Sigma_5^{\text{term}}[\mathcal{K}_+] = \emptyset \implies \neg \text{RealizationLicensed}(\mathfrak{C}_{v,\text{term}}^{(5)}, T_{v,\text{term}}^{(5)}).
 \end{aligned} \tag{73}$$

Table 6: Load-bearing falsifiers.

Claim	Faithful falsifier	Consequence for scope
PCR–4 carrier multiplicity	Show one benchmark fails an imported simultaneous constraint or that the parent carrier traces are equivalent.	PCR–3/PCR–4.
Universal internal lifting	Give a PCR–4 survivor whose finite complete witness packet admits no complete E_5/J_5 -positive maximal factorization assignment.	PCR–5 internal occupancy analysis.
Carrier-semantic descent	Produce two carrier-equivalent internal orbits that the terminal projection separates, or two carrier-inequivalent parents that the frozen carrier equivalence identifies.	PCR–3 identity / theorem 16.4.
Bounded carrier multiplicity	Supply an already-present same-target selector eliminating one benchmark carrier class or a lawful carrier-trace isomorphism.	Cross-corpus disposition.
Terminal role typing	Show the separately declared terminal target has a role architecture other than the one complete-carrier slot.	Replace $T_{v,\text{term}}^{(5)}$.
Terminal eligibility	Show a claimed E_5^{term} edge lacks target-role or carrier-provenance formation standing.	Re-evaluate terminal E .
Terminal compatibility	Show a claimed J_5^{term} -positive carrier has no complete internal compatible occupancy witness.	Re-evaluate the terminal handoff or internal J_5 .
Terminal live residue	Exhibit a positive terminal preassignment omitted from the union, or a listed live carrier with no positive preassignment.	Recompute $\text{Residue}_{\text{term}}^{\text{live}}$.

Claim	Faithful falsifier	Consequence for scope
Terminal ROC	Produce a bijection from the singleton terminal role to the complete live carrier residue, or invalidate role-realization completeness.	Re-evaluate the terminal certificate.

26.1 Excluded inference patterns

The following non-implications delimit the logical scope of the theorem:

$$\text{common cone} \not\Rightarrow \text{one normalized shared cell}, \quad (74)$$

$$\Sigma_5^{\text{int}} \neq \Sigma_5^{\text{term}}, \quad (75)$$

$$\text{admissible factorizations} \not\Rightarrow \text{standing-valid common refinement}, \quad (76)$$

$$\text{internal complete orbit} \not\Rightarrow \text{terminal live residue without } E_5^{\text{term}} \text{ and } J_5^{\text{term}}. \quad (77)$$

In addition, factorization difference alone does not imply terminal carrier difference:

$$\text{incomparable maximal factorizations over one carrier} \not\Rightarrow \text{distinct terminal carrier residues.}$$

26.2 Conditions for revising the conclusion

The terminal carrier residue can change only if the target is explicitly enlarged, a premise in (73) is shown false, a carrier is eliminated by a same-target admissibility theorem, or a lawful carrier-trace equivalence is established. A post hoc preference is not a same-target mathematical revision.

27 Outlook: upper closure and UV-sensitive target extensions

The PCR-5 result leaves two related objects for subsequent analysis:

$$\mathfrak{D}_5^{\text{int}} = \bigsqcup_{[x] \in \mathfrak{G}_\nu^{(4)}} \mathfrak{F}_5([x]), \quad \mathfrak{G}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+] = \bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]), \quad (78)$$

with $|\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]| \geq 2$, $|\mathfrak{G}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2$, and terminal over-residue. The first object retains factorization-sensitive occupancy architecture; the second is the factorization-insensitive carrier-semantic residue relevant to physical carrier selection.

27.1 Two successor problems

The first problem is an *upper-closure theorem*: classify the carrier-semantic PCR-5 residue sufficiently to decide whether the lower bound is exact, while keeping internal factorization multiplicity distinct from carrier multiplicity. This is not required for the PCR-5 non-singleton result.

The second problem is an enlarged UV-sensitive target. Candidate coordinates include direct heavy-neutral-lepton standing, correlated nonunitarity, charged-lepton-flavor violation, collider or cosmological residues, dimension-six effects, or a fully specified $B - L$ /global-form/breaking architecture. Each requires its own target certificate, null-result semantics, candidate completeness, and lawful quotient.

Theorem 27.1 (No PCR-6 backflow). *A property derived only in a successor target cannot change $\mathfrak{D}_5^{\text{int}}$, $\mathfrak{G}_{\nu, \text{car}}^{(5)}$, $\mathfrak{G}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$, or the terminal over-residue theorem without identifying a false PCR-5 premise or explicitly replacing the target certificate.*

Proof. PCR-5 fixes its source classes, factorization-sensitive internal construction, carrier projection, terminal eligibility and compatibility relations, and terminal role architecture before any successor observable is introduced. A later target may define a new restricted domain but cannot retroactively alter these fixed objects. $\square$

Outlook

Any successor analysis therefore begins from a cross-corpus-stable carrier-semantic residue with at least two classes and a terminal ROC selector gap. It may either close the carrier upper bound or introduce a genuinely new standing-bearing selector through an explicitly enlarged target.

28 Conclusion

PCR-5 asks whether the exact PCR-4 carrier survivor residue closes to one compatible physical carrier realization after role occupancy is resolved. The analysis separates three distinct levels: internal factorization, carrier-semantic identity, and terminal Role-Occupancy Closure.

First, every PCR-4 survivor lifts to at least one complete internally compatible role-bearing factorization. The one-block core factorization establishes nonemptiness of the admissible-factorization poset; finite maximality supplies at least one maximal element; every retained factor block has nonempty role-standing signature; and the complete factorized diagram passes E_5 and J_5 . Thus

$$\mathfrak{D}_5^{\text{int}} = \bigsqcup_{[x] \in \mathfrak{S}_v^{(4)}} \mathfrak{F}_5([x]), \quad \mathfrak{F}_5([x]) \neq \emptyset, \quad |\mathfrak{D}_5^{\text{int}}| \geq |\mathfrak{S}_v^{(4)}| \geq 2. \quad (79)$$

Compatible occupancy therefore universally lifts the PCR-4 carrier residue.

Second, carrier resolution is deliberately coarser than the factorization-sensitive internal classifier. Alternative maximal factorizations over one parent carrier are quotiented by the carrier projection:

$$\mathfrak{S}_{v,\text{car}}^{(5)} := \text{im}(\bar{\pi}_{\text{car}}) = \mathfrak{S}_v^{(4)}. \quad (80)$$

The law-only Weinberg point and the constructive full-rank active SM + $3\nu_R$ point remain distinct at this factorization-insensitive resolution because their fixed parent carrier traces are inequivalent. After the bounded corpus intersection,

$$\mathfrak{S}_{v,\text{car}}^{(5)}[\mathcal{K}_+] := \bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]), \quad |\mathfrak{S}_{v,\text{car}}^{(5)}[\mathcal{K}_+]| \geq 2. \quad (81)$$

Low-energy matching may identify κ , propagation kernels, or numerical residues, but not primitive nodes, source incidence, retained-resolution continuation, or failure domains.

Third, terminal Role-Occupancy Closure is evaluated with the complete carrier-resolution type chain. The terminal target has the fixed role set

$$R_5^{\text{term}} = \{r_{\text{car}}(T_v^{\text{osc}})\}.$$

Every carrier survivor receives a same-certificate eligibility edge E_5^{term} , and terminal compatibility J_5^{term} is positive precisely when that carrier possesses at least one complete internal compatible occupancy witness at the relevant certificate level. Hence

$$\text{Residue}_{\text{term}}^{\text{live}} = \mathfrak{S}_{v,\text{car}}^{(5)} = \mathfrak{S}_v^{(4)}, \quad \text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] = \mathfrak{S}_{v,\text{car}}^{(5)}[\mathcal{K}_+]. \quad (82)$$

Because both live residues have cardinality at least two while $|R_5^{\text{term}}| = 1$, no compatible bijective role-covering assignment exists:

$$\Sigma_5^{\text{term}} = \emptyset, \quad \Sigma_5^{\text{term}}[\mathcal{K}_+] = \emptyset. \quad (83)$$

The terminal certificate is role-realization complete at the declared carrier resolution, so Role-Occupancy Closure yields

$$\neg \text{RealizationLicensed}(\mathfrak{C}_{v,\text{term}}^{(5)}, T_{v,\text{term}}^{(5)}). \quad (84)$$

The result is a strict lower bound, not an exact two-carrier theorem. A future singleton theorem would require one of three mathematically distinct developments: failure of an upstream premise, a lawful carrier-trace equivalence, or an explicitly enlarged target carrying new UV-sensitive standing. Alternative factorization, low-energy matching, or post hoc preference cannot by themselves supply such a collapse.

Theorem summary

$$\begin{aligned}
& \forall x \in \mathfrak{S}_\nu^{(4)} : \Sigma_5^{\text{int}}(x) \neq \emptyset, \quad \mathfrak{S}_{\nu, \text{car}}^{(5)} = \text{im}(\bar{\pi}_{\text{car}}) = \mathfrak{S}_\nu^{(4)}, \\
& |\mathfrak{S}_{\nu, \text{car}}^{(5)}| \geq 2, \quad q_W, q_{3N} \in \mathfrak{D}_5^{\text{int}}[\mathcal{K}_+], \quad \bar{\pi}_{\text{car}}(q_W) \neq \bar{\pi}_{\text{car}}(q_{3N}), \\
& \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+] = \bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]), \quad |\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2, \\
& E_5^{\text{term}} + J_5^{\text{term}} \implies \text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] = \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+], \quad |\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+]| \geq 2 > |R_5^{\text{term}}| = 1, \\
& \Sigma_5^{\text{term}}[\mathcal{K}_+] = \emptyset, \quad \neg \text{RealizationLicensed}(\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)}).
\end{aligned}$$

A Notation, object typing, and target structure

This appendix fixes the three resolutions used throughout the manuscript: internal factorization-sensitive occupancy, carrier-semantic realization, and terminal Role-Occupancy Closure.

Table 7: Principal notation.

Symbol	Meaning
$\mathfrak{S}_\nu^{(4)}$	PCR-4 carrier-semantic survivor quotient at the frozen simultaneous target.
$T_{\nu, \text{int}}^{(5)}$	Internal PCR-5 target: complete role-bearing factorization architectures inside one PCR-4 survivor.
$\mathcal{G}_5$	Independently frozen internal grammar of target, source, witness, normalization, role-core, factorization, boundary, continuation, parameter, failure, and lawful-redescription coordinates.
$\mathcal{N}_5(x)$	Witness-preserving irreducible normal forms reachable from the complete PCR-4 support.
$\text{Core}_5(N)$	Source-constitutive closure of all mandatory role/interface witness paths.
$\text{Aux}_5(N)$	Parent-carrier and standing trace not automatically promoted to occupancy blocks.
$\text{Fact}_5(N)$	Finite poset of admissible role-bearing factorizations.
$\text{MaxFact}_5(N)$	Maximal admissible factorizations under lawful refinement; uniqueness is not assumed.
$\text{RSig}_5(\omega; N)$	Nonempty role-standing signature required of every occupancy block.
Σ_5^{int}	Complete factorization-sensitive internal assignment space.
Γ_5^{int}	Lawful internal presentation groupoid preserving parent carrier trace and the selected factorization-sensitive invariant.
$\mathfrak{D}_5^{\text{int}}$	Factorization-sensitive internal orbit object $\Sigma_5^{\text{int}}/\Gamma_5^{\text{int}}$.
$\mathfrak{F}_5([x])$	Nonempty internal orbit fiber above carrier class $[x]$.
$\bar{\pi}_{\text{car}}$	Carrier projection from internal orbits to the frozen carrier-semantic quotient.
$\mathfrak{S}_{\nu, \text{car}}^{(5)}$	Native PCR-5 carrier-semantic residue $\text{im}(\bar{\pi}_{\text{car}}) = \mathfrak{S}_\nu^{(4)}$.
$\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]$	Bounded-support factorization-sensitive internal orbit object.

Symbol	Meaning
$\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]$	Bounded carrier-semantic residue $\bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+])$, with cardinality at least two.
$T_{\nu, \text{term}}^{(5)}$	Separately declared carrier-resolution terminal target, fixed before residue inspection.
$R_5^{\text{term}} = \{r_{\text{car}}(T_{\nu}^{\text{osc}})\}$	One complete-carrier role specified by the terminal target architecture.
E_5^{term}	Terminal same-certificate eligibility relation from carrier-semantic residue to the complete-carrier role.
J_5^{term}	Terminal compatibility predicate; positive when a carrier class has a complete internally compatible occupancy witness.
$\Sigma_5^{\text{pre, term}}$	Injective pointwise eligible, terminal-compatible preassignments.
$\text{Residue}_{\text{term}}^{\text{live}}$	ROC live carrier residue, the union of images of all positive terminal preassignments.
Σ_5^{term}	Compatible bijective role-covering assignment set from R_5^{term} onto the complete terminal live residue.
I_5^{term}	Terminal assignment invariant containing the fixed target, role set, live residue, eligibility, compatibility, carrier standing, boundary, continuation, and failure data.
Γ_5^{term}	Certificate-lawful terminal automorphism group $\text{Aut}(I_5^{\text{term}})$.
OverResidue	ROC diagnosis $ \text{Residue}_{\text{term}}^{\text{live}} > R_5^{\text{term}} $.

A.1 Identity hierarchy

The relevant implication chain is

$$\begin{aligned}
 \text{factorization-sensitive internal identity} &\implies \text{parent carrier identity} \\
 &\implies \text{low-energy matching class} \\
 &\implies \text{interface equality.}
 \end{aligned} \tag{85}$$

Neither converse is automatic. The terminal target intentionally resolves only to the parent carrier-semantic level.

A.2 Typing rules

1. $\text{Core}_5(N)$ is generated from mandatory witness paths; auxiliary trace is not automatically an occupancy block.
2. Incomparable maximal factorizations are retained internally; no common refinement is inferred.
3. Internal factorization difference alone does not imply distinct terminal carrier residue.
4. The terminal carrier quotient is the carrier projection, not $\mathfrak{D}_5^{\text{int}}$ itself.
5. The terminal role architecture is declared before residue inspection; it is not inferred from one target instance or from endpoint cardinality.
6. E_5^{term} and J_5^{term} are separately formed before $\Sigma_5^{\text{pre, term}}$ and $\text{Residue}_{\text{term}}^{\text{live}}$.
7. Internal E_5/J_5 supplies the witness for terminal compatibility but is not silently substituted for terminal E/J .
8. No canonical section $\mathfrak{S}_{\nu}^{(4)} \rightarrow \mathfrak{D}_5^{\text{int}}$ is assumed.

B Normalization, maximal factorization, and pattern catalogues

B.1 Reduction catalogue

Table 8: Witness-preserving normalization rules.

Rule	Permitted operation	Mandatory side condition
N1	Delete target-inert support.	Every mandatory witness path, continuation/failure record, and parent carrier coordinate is preserved.
N2	Remove a duplicate presentation.	An imported lawful redescription identifies both presentations.
N3	Contract a constitutive chain.	The internal vertex has no independent generated coordinate.
N4	Replace a witness cone by an isomorphic presentation.	All legs, commutativities, ownership, and standing are preserved.
N5	Delete a source-null contribution at the fixed point.	Pointwise nullity is proved and no mandatory witness depends on it.

Every rule decreases the well-founded measure of lemma 6.1. Confluence is not assumed.

B.2 Admissible factorization clauses

Table 9: Admissible role-bearing factorization clauses.

ID	Clause	Pass condition
F1	Full core coverage	Blocks partition all of $\text{Core}_5(N)$.
F2	Source coherence	Every block is source-convex in the core graph.
F3	Reconstruction	Blocks and retained edges reconstruct the complete role/interface witness diagram.
F4	Role standing	Every block has nonempty RSig_5 .
F5	Activity	Every block is non-null at the fixed source point.
F6	No duplicates	Distinct blocks are not presentation duplicates.
F7	No repair	Deletion loss cannot be repaired by another point, carrier, or unregistered interaction.
F8	Parent preservation	Complete carrier and auxiliary trace remain attached.

The one-block factorization establishes nonemptiness. The finite partition lattice supplies maximal admissible refinements. Maximal elements need not be unique or mutually comparable.

B.3 Pattern catalogue

Table 10: Patterns derived from one maximal factorization.

Pattern	Block condition	Role-incidence condition	Assignment condition
P_{sh}	Exactly one factor block.	Its role-standing signature contains both roles.	One address maps to the unique block.
P_{nd}	At least two blocks.	Distinct role kernels are linked by directed constitutive nesting without irreducibly branching joint standing.	Every block and nesting edge is retained.
P_{dist}	At least two blocks and not nested-distinct.	At least one mandatory role witness or its standing is borne only by the joint factor graph.	Every block and joint edge is retained.

B.4 Excluded factorization forms

Table 11: Rejected factorization and assignment proposals.

Proposal	Reason for rejection
Common cone forced to one block	Co-instantiation does not prove irreducibility.
Common refinement of admissible factorizations	Role standing need not be hereditary under intersection.
Auxiliary-only object counted as a block	Complete trace relevance is not role occupancy.
Block with empty role signature	Violates F4 and cannot receive an occupancy address.
One maximal factorization discarded	Incomparable maximal elements remain live.
Nonmaximal factorization advertised as exact	Omits a proper admissible refinement.
Pattern selected before factorization	Reverses dependency order.
Proper-subset codomain	Leaves a role-bearing factor block unused.
Decoupling used as block criterion	Admissible factors may remain coupled by retained constitutive maps.
Weak-basis labels used as invariant blocks	Basis names do not establish physical standing.

C Eligibility certificate

C.1 Clause ledger

Table 12: The nine E_5 clauses.

ID	Clause	Pass condition
E1	Role-standing adequacy	The block has nonempty RSig_5 and supports every recorded role.
E2	Positive witness	Its contribution factors through the PCR-4 role/channel/response packet.
E3	Deletion sensitivity	Removing it without repair destroys an assigned witness or standing path.
E4	Source activity	It is non-null at the fixed PCR-3 source point.
E5	Target participation	It lies on a mandatory source-to-interface role path.
E6	Derived-factor typing	Address, block, incidence, and edge type agree in $(G_{N,F}, I_{N,F})$.
E7	Trace sufficiency	Every generated evaluation coordinate is defined.
E8	One-survivor standing	No coordinate is borrowed from another carrier point.
E9	Lawful invariance	The disposition is invariant under lawful presentation isomorphism.

C.2 Benchmark dispositions

Table 13: Benchmark eligibility certificates.

Witness	Status	Load-bearing evidence
LAW-W / every block of F_W	PASS	F_W is maximal within the admissible role-bearing factorization poset; every block has nonempty role-standing signature and lies on the complete law-only source-to-interface witness.
LOCAL-3N / every block of F_{3N}	PASS	The constructive full-rank point has a complete active source-to-visible role core; every block of the selected maximal factorization satisfies F1–F8.

C.3 Failure dispositions

Table 14: Representative E_5 failures.

Proposal	Disposition
Auxiliary trace promoted to a block without role standing	E1 fails.
Block retained with empty RSig_5	E1 and E3 fail.
Block deleted and repaired at another parameter point	E3 and E8 fail.
Source-null singlet retained	E4 fails.
Weak-basis label treated as invariant block	E6 and E9 fail.

Proposal	Disposition
Proper subset of a maximal factorization assigned	Full coverage and J_5 fail.

D Compatibility certificate

Table 15: The ten J_5 clauses.

ID	Clause	Pass condition
J1	Full factor coverage	Every address and factor block is used exactly once.
J2	One parent	All blocks and auxiliary traces restrict one survivor.
J3	One parameter point	All evaluations occur at the same retained point.
J4	Source reconstruction	Every inter-block constitutive map is retained.
J5	Ordered roles	$r_\Delta \succ r_{\triangleright\triangleleft}$ holds on the complete diagram.
J6	Common channel/response	PCR-4 common witnesses are used.
J7	Common ledgers	Boundary, continuation, charge, anomaly, vacuum, EFT, and failure data agree.
J8	No duplication	Support is not duplicated across addresses.
J9	Common cone	The complete factorized witness admits the preserved PCR-4 cone.
J10	Core/auxiliary typing	Auxiliary data retain parent incidence but are not counted as additional occupancy blocks.

D.1 Benchmark compatibility

Table 16: Benchmark J_5 dispositions.

Assignment	Factorization status	Compatibility evidence
z_W	Some $F_W \in \text{MaxFact}_5(N_W)$	Full core coverage, one parent and point, complete law-only witness reconstruction, one common cone, and one consistency/failure ledger.
z_{3N}	Some $F_{3N} \in \text{MaxFact}_5(N_{3N})$	Full core coverage, all retained constitutive maps, one full-rank source point, one common cone, and one heavy-to-visible consistency/failure ledger.

No exact factor count is part of either compatibility certificate.

E Internal groupoid, carrier descent, and terminal ROC

E.1 Internal groupoid generators

Allowed generators preserve the complete factorization-sensitive internal object

$$(x, N, F, \text{Core}_5(N), \text{Aux}_5(N), G_{N,F}, I_{N,F}, P_{N,F}, \sigma, E_5, J_5, \tau_{\text{asg}}^{(5)}).$$

They include lawful address relabeling, basis/gauge/source-diagram isomorphism, reparameterization within one stratum, and isomorphic normal-form or maximal-factorization presentation. They exclude

common-refinement replacement, factor deletion, interface-only identification, EFT matching as identity, and cross-carrier morphisms.

E.2 Carrier-indexed decomposition and coarser terminal quotient

The carrier projection is orbit constant and yields

$$\mathfrak{D}_5^{\text{int}} = \bigsqcup_{[x] \in \mathfrak{S}_\nu^{(4)}} \mathfrak{F}_5([x]), \quad \mathfrak{F}_5([x]) \neq \emptyset.$$

The terminal target does not preserve the factorization graph as an independent carrier coordinate. It applies the coarser equivalence

$$q \equiv_{\text{term}} q' \iff \bar{\pi}_{\text{car}}(q) = \bar{\pi}_{\text{car}}(q'),$$

so

$$\mathfrak{D}_5^{\text{int}} / \equiv_{\text{term}} \cong \mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_\nu^{(4)}.$$

Thus several internal factorization orbits over one parent do not manufacture multiple terminal carrier residues.

E.3 Terminal eligibility and compatibility

The terminal candidate residue is $\mathfrak{S}_\nu^{(4)}$, with

$$E_5^{\text{term}}([x], r_{\text{car}}(T_\nu^{\text{osc}})) = 1 \quad \text{for every } [x] \in \mathfrak{S}_\nu^{(4)}.$$

For $\tau_{[x]}(r_{\text{car}}(T_\nu^{\text{osc}})) = [x]$,

$$J_5^{\text{term}}(\tau_{[x]}) = 1 \iff \mathfrak{F}_5([x]) \neq \emptyset.$$

The bounded version replaces the native fiber by its intersection with $\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]$. Consequently

$$\text{Residue}_{\text{term}}^{\text{live}} = \mathfrak{S}_{\nu, \text{car}}^{(5)} = \mathfrak{S}_\nu^{(4)}, \quad \text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] = \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+].$$

At terminal resolution the lawful invariant and automorphism group may be written

$$I_5^{\text{term}} = (\mathfrak{C}_{\nu, \text{term}}^{(5)}, T_{\nu, \text{term}}^{(5)}, R_5^{\text{term}}, \text{Residue}_{\text{term}}^{\text{live}}, E_5^{\text{term}}, J_5^{\text{term}}, \chi_{\text{car}}, \mathcal{B}_{\text{term}}, \mathcal{C}_{\text{term}}, \mathcal{F}_{\text{term}}), \quad \Gamma_5^{\text{term}} = \text{Aut}(I_5^{\text{term}}),$$

with the bounded certificate using the corresponding restriction $\Gamma_5^{\text{term}}[\mathcal{K}_+]$. The compatible role-covering assignment set is empty because $|R_5^{\text{term}}| = 1$ while both live carrier residues have cardinality at least two; consequently the corresponding lawful-orbit quotients are empty as well.

F Familywise factorization-fiber atlas

The supplementary family atlas records the following invariant statement for every PCR–4 survivor family:

$$\text{surviving point} \implies \mathcal{N}_5(x) \neq \emptyset \implies \text{MaxFact}_5(N) \neq \emptyset \implies \Sigma_5^{\text{int}}(x) \neq \emptyset.$$

No row assigns a universal factor count or pattern to a family. Pure loci excluded by PCR–4 remain empty; surviving hybrid, composite, solitonic, hidden, and environment-bearing points receive factorization fibers only at their admitted loci.

G Explicit benchmark factorization witnesses

G.1 Law-only Weinberg realization

Table 17: Weinberg benchmark witness packet.

Coordinate	Certified value
Parent	Minimal-SM field carrier with active dimension-five Weinberg source.
Primitive extra nodes	None.
Fixed coefficient	$\kappa_* = UDU^T$ with target-admissible mixing and response separation.
Role core	Complete source-to-interface law, propagation, channel, response, relation, and common-cone paths.
Normal form	Any $N_W \in \mathcal{N}_5(x_W)$.
Maximal factorization	Any $F_W \in \text{MaxFact}_5(N_W)$; exact block count not claimed.
Assignment	Tautological bijection onto all blocks of F_W .
E_5/J_5 status	Every block passes E_5 ; complete diagram passes J_5 .
Orbit	$q_W = [z_W]_{\Gamma_5^{\text{int}}}$.

G.2 Active three-singlet realization

Table 18: Three-singlet benchmark witness packet.

Coordinate	Certified value
Parent	SM + $3\nu_R$ with rank $Y_\nu = 3$ and $\det M_R \neq 0$.
Primitive extra nodes	Three active retained singlet nodes.
Constructive point	$Y_\nu = UD^{1/2}M_R^{-1/2}$, hence $Y_\nu M_R^{-1}Y_\nu^T = UDU^T = \kappa_*$.
Role core	Complete active heavy-source-to-visible role/interface witness paths.
Normal form	Any $N_{3N} \in \mathcal{N}_5(x_{3N})$.
Maximal factorization	Any $F_{3N} \in \text{MaxFact}_5(N_{3N})$; basis-covariant finer factorizations are retained.
Assignment	Tautological bijection onto all blocks of F_{3N} .
E_5/J_5 status	Every block passes E_5 ; complete diagram passes J_5 .
Orbit	$q_{3N} = [z_{3N}]_{\Gamma_5^{\text{int}}}$.

The benchmarks can share

$$\pi_{\text{LE}}(q_W) = \pi_{\text{LE}}(q_{3N}) = \kappa_*$$

while remaining carrier inequivalent. The distinction is in primitive nodes, attachment incidence, ultraviolet continuation, and failure domain; it does not depend on factor count.

H Compact theorem package and dependency matrix

H.1 Theorem chain

$$\begin{aligned}
x \in \mathfrak{S}_\nu^{(4)} &\implies \text{JointSupp}_5(x) \text{ finite and complete} \\
&\implies \mathcal{N}_5(x) \neq \emptyset \\
&\implies \text{Core}_5(N) \neq \emptyset \\
&\implies \text{MaxFact}_5(N) \neq \emptyset \\
&\implies \Sigma_5^{\text{int}}(x) \neq \emptyset \\
&\implies \mathfrak{F}_5([x]) \neq \emptyset.
\end{aligned}$$

Therefore

$$\mathfrak{D}_5^{\text{int}} = \bigsqcup_{[x] \in \mathfrak{S}_\nu^{(4)}} \mathfrak{F}_5([x]), \quad \mathfrak{S}_{\nu, \text{car}}^{(5)} = \text{im}(\bar{\pi}_{\text{car}}) = \mathfrak{S}_\nu^{(4)}, \quad |\mathfrak{S}_{\nu, \text{car}}^{(5)}| \geq 2.$$

After bounded corpus restriction,

$$q_W, q_{3N} \in \mathfrak{D}_5^{\text{int}}[\mathcal{K}_+], \quad \bar{\pi}_{\text{car}}(q_W) \neq \bar{\pi}_{\text{car}}(q_{3N}), \quad |\mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+]| \geq 2.$$

The terminal sequence is

$$\begin{array}{ccccc}
R_5^{\text{term}} & \xrightarrow{\text{fixed target architecture}} & E_5^{\text{term}} & \xrightarrow{\text{carrier provenance}} & J_5^{\text{term}}, \\
J_5^{\text{term}} & \xrightarrow{\text{internal compatible lift}} & \Sigma_5^{\text{pre, term}} & \xrightarrow{\text{union of images}} & \text{Residue}_{\text{term}}^{\text{live}}.
\end{array}$$

It yields

$$\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+] = \mathfrak{S}_{\nu, \text{car}}^{(5)}[\mathcal{K}_+], \quad |\text{Residue}_{\text{term}}^{\text{live}}[\mathcal{K}_+]| \geq 2 > |R_5^{\text{term}}| = 1, \quad \Sigma_5^{\text{term}}[\mathcal{K}_+] = \emptyset,$$

and therefore unique realization is not licensed.

Table 19: Dependency matrix.

Result	Depends on	Does not assume
Normal-form existence	Finite support and well-founded reduction measure	Confluence or preferred normal form.
Role-core/maximal-factorization existence	Preserved mandatory witness paths, one-block admissibility, finite refinement poset	Common refinement or unique factorization.
Universal internal lift	Factor-block E_5 , complete-diagram J_5 , one parent	Canonical section or model preference.
Factorization-sensitive internal quotient	Declared $\mathcal{G}_5$ -generated decomposition uses	Exact terminal carrier semantics.
Carrier-semantic descent	Frozen carrier projection and nonempty fibers	Physical inequivalence of incomparable factorizations.
Bounded carrier multiplicity	Two carrier-distinct bounded-support benchmark images	Exact total carrier count.
Terminal role set	Separately declared one-role carrier-resolution target	One-target-instance implies one role; residue cardinality.
Terminal eligibility	Same-target carrier typing and provenance	Internal E_5 substituted by fiat.

Result	Depends on	Does not assume
Terminal compatibility	Existence of complete internal E_5/J_5 -positive witness	Pointwise eligibility alone.
Terminal live residue	Union of all positive terminal preassignment images	Direct use of $\mathfrak{D}_5^{\text{int}}$ as terminal residue.
ROC capstone	Role-realization-complete terminal certificate and live carrier multiplicity	Factorization-overcount or hidden selector.

I Source provenance and dependency lock

I.1 Primary dependencies

Table 20: Primary source register.

Stage	Imported object	PCR-5 use
PCR-1	Target-fixed role necessity	Fixes the two ordered physical roles without selecting a carrier.
PCR-2	Complete physical interface	Fixes target-facing channel and response requirements.
PCR-3	Source-first carrier universe	Supplies parent carrier traces and lawful same-regime comparison classes.
PCR-4	Simultaneous survivor quotient	Supplies complete one-source common-cone witness packets and at least two carrier-inequivalent survivors.
ROC	General role-occupancy closure theorem	Applied only to the fixed terminal one-role target after internal quotienting.

I.2 Added proof-level support

Table 21: Added source classes and their bounded use.

Source class	Disposition
SM carrier and cardinality ARC	Sharpens generation role count and fixed-load scope; does not identify generation roles with retained singlet primitives.
Full SYM reference ARC	Keeps Weinberg and $\text{SM} + 3\nu_R$ realizations in distinct target fibers and types matching as inter-target EFT transport.
Neutrino support corpus	Fixes low-energy character and numerical residues without selecting ultraviolet origin.
Realized-interior, bookkeeping, charge-lattice, and RG papers	Supply common or scope-guarded structure and block matching-as-identity.
SR1–SR7	Closes a common mass-pre-dynamical gauge skeleton, leaving neutrino mechanism downstream.

Source class	Disposition
EX1–EX9	Blocks promotion from quotient, numerical, transport, or extraction exactness to raw carrier-origin singletonhood.

The supplementary source register records eighty PDFs spanning 5,099 pages, with seventy-nine unique document payloads. It establishes provenance and bounded coverage but does not substitute for the semantic proofs of normalization, eligibility, compatibility, fiber separation, or terminal Role-Occupancy Closure.

I.3 Dependency lock

Downstream constraints may restrict the internal orbit object only when they are well typed on the unchanged PCR–5 target. They may not be back-propagated to rewrite an upstream source class, delete a PCR–4 survivor by a different fixed-load theorem, or enlarge the terminal role set after the live residue is known.

J Adversarial consistency checks

A supplementary family of seventy-one negative controls tests common violations of the declared dependency and typing structure. The controls are grouped below by logical class; their purpose is to demonstrate robustness of the formal representation under targeted perturbations, not to replace the mathematical proofs.

Table 22: Adversarial consistency classes M01–M64.

Range	Logical class	Required preservation
M01–M34	Role typing, activity, ownership, carrier identity, target scope, cross-corpus projection, exactness, and certificate governance	Preserve the fixed target, source point, carrier trace, lower-bound discipline, and no-selector rules.
M35–M52	Full codomain, derived pattern, common-cone discipline, normalization, carrier fibers, two-level typing, active benchmark, grammar boundary	Preserve complete internal assignments and prohibit direct internal-object substitution into ROC.
M53–M64	Role-bearing cores, maximal factorizations, common-refinement prohibition, basis covariance, target-before-count, factor-count-neutral benchmark separation	Preserve every role-bearing maximal factorization without promoting unproved refinements or factor counts into endpoint selection.

Table 23: Terminal consistency checks M65–M71.

ID	Perturbation	Rejection condition
M65	Treat every internal factorization-sensitive orbit as a distinct terminal carrier residue.	Terminal residue is the carrier projection; factorization variants over one parent collapse.
M66	Omit the terminal eligibility relation and map the terminal role directly to carrier candidates.	Every terminal preassignment must pass the explicitly formed E_5^{term} edge.
M67	Omit terminal compatibility or replace it by pointwise eligibility.	J_5^{term} requires a complete internal compatible occupancy witness.

ID	Perturbation	Rejection condition
M68	Substitute internal E_5/J_5 directly for terminal $E_5^{\text{term}}/J_5^{\text{term}}$.	The internal-to-terminal handoff theorem is required because the predicates have different types.
M69	Infer one terminal role merely from the fact that there is one target instance.	The one role is fixed by the terminal target architecture, not instance cardinality.
M70	Set the terminal live residue equal to $\mathfrak{D}_5^{\text{int}}$ or $\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+]$.	Live residue is the union of positive terminal preassignment images and equals the carrier-semantic projection.
M71	Treat incompatible maximal factorizations over one carrier as automatically distinct physical carrier realizations.	Factorization difference alone has no terminal standing; parent carrier trace controls terminal identity.

The supplementary checks confirm that the encoded data remain consistent with the theorem architecture. They do not provide an independent proof of the semantic results.

K Reproducibility and internal consistency

The supplementary source archive provides structured records for the target certificate, normalization grammar, role-core and factorization data, benchmark witnesses, family fibers, the factorization-sensitive internal groupoid, carrier-semantic terminal quotient, terminal eligibility and compatibility, the endpoint certificate, and the bounded cross-corpus analysis. These records are intended to make the finite representation of the theorem independently inspectable.

K.1 Consistency conditions

The supplementary checks verify that:

1. internal and terminal targets are separately typed and occur in the correct dependency order;
2. every normal form has a nonempty role-bearing core and every role core has at least one maximal admissible factorization;
3. every occupancy block has nonempty role-standing signature, while auxiliary-only data remain trace annotations;
4. every internal assignment covers an entire selected maximal factorization and no common refinement is inferred;
5. every PCR-4 survivor locus has a nonempty internal assignment fiber;
6. both explicit benchmarks have complete assignments and their carrier-trace separation is factor-count independent;
7. the internal quotient is factorization-sensitive, whereas terminal carrier semantics are obtained by the coarser carrier projection $\bar{\pi}_{\text{car}}$;
8. maximal-factorization variants over one carrier are not overcounted as distinct terminal residues;
9. the terminal role architecture is fixed before residue inspection;
10. terminal E_5^{term} and J_5^{term} are defined separately and every terminal preassignment passes both;
11. the native carrier-semantic live residue equals $\mathfrak{S}_v^{(4)}$, while the bounded live residue is $\bar{\pi}_{\text{car}}(\mathfrak{D}_5^{\text{int}}[\mathcal{K}_+])$ and contains at least two carrier classes;

12. no terminal role-covering bijection exists, so the ROC diagnosis is over-residue;
13. all twenty-nine bounded-corpus selector classes have explicit dispositions; and
14. all seventy-one adversarial consistency checks are rejected as required.

K.2 Scope of computational verification

The accompanying consistency checker can detect malformed factorization clauses, empty role signatures, auxiliary-as-cell errors, common-refinement promotion, factorization overcount at terminal resolution, omitted terminal eligibility or compatibility, silent substitution of internal E_5/J_5 for terminal E/J , target-role backfitting, cross-carrier morphisms, missing fibers, and inconsistent corpus counts. It does not establish the physical truth of PCR–1 through PCR–4 or semantic completeness beyond the fixed grammar and bounded reference corpus. The finite records retain their source schema; their consistency checks do not formalize the primitive-generation receipt in Proposition 23.9. That receipt is a manuscript theorem with its own source interpretation and proof.

K.3 Reproducibility materials

The source archive includes scripts for consistency verification, bibliography-resolved compilation, PDF preflight, and checksum generation. These materials permit independent reconstruction of the published document and verification of the supplementary records. They support provenance and reproducibility only; the mathematical conclusions rest on the theorem chain in the main text.

L Nonclaims and exact scope boundary

Nonclaim L.1 (No exact two-carrier theorem). *PCR–5 proves $|\mathfrak{S}_{\mathbf{v},\text{car}}^{(5)}[\mathcal{K}_+]| \geq 2$. It does not prove that exactly two carrier-semantic classes survive.*

Nonclaim L.2 (No terminal multiplicity from factorization count). *Several incomparable maximal factorizations or internal orbits over one parent carrier do not, by themselves, define several terminal carrier residues. The terminal quotient collapses them through $\bar{\pi}_{\text{car}}$.*

Nonclaim L.3 (No canonical factorization). *The common PCR–4 cone does not imply one block, and admissible factorizations are not replaced by an unproved common refinement.*

Nonclaim L.4 (No exact benchmark pattern theorem). *The lower bound does not require the Weinberg assignment to be one-block or the three-singlet assignment to have exactly two distributed blocks.*

Nonclaim L.5 (No terminal role from instance cardinality). *The one terminal role is a declared field of the carrier-resolution target architecture. The manuscript does not infer that every one-target-instance problem has one role.*

Nonclaim L.6 (No silent internal-to-terminal handoff). *Internal E_5/J_5 are not terminal $E_5^{\text{term}}/J_5^{\text{term}}$. They provide the proved co-instantiation witness used by the separately defined terminal compatibility predicate.*

Nonclaim L.7 (No direct internal ROC substitution). *Σ_5^{int} and $\mathfrak{D}_5^{\text{int}}$ are internal factorization-sensitive objects. ROC is applied only after carrier-semantic descent and explicit terminal eligibility, compatibility, and live-residue formation.*

Nonclaim L.8 (No denial of realization existence). *Terminal over-residue means that more than one compatible carrier-semantic residue survives; it does not mean that no realization exists.*

Nonclaim L.9 (No ultraviolet identity from matching). *The relation $\kappa_{\text{eff}} = Y_{\mathbf{v}} M_{\mathbf{R}}^{-1} Y_{\mathbf{v}}^T$ may reproduce the same low-energy coefficient without identifying primitive nodes, source incidence, continuation, or failure domains.*

Nonclaim L.10 (No unrestricted semantic exhaustion). *Internal completeness is relative to $\mathcal{G}_5$; terminal completeness is relative to the declared carrier-resolution target and frozen/bounded certificates. Future target updates may add new coordinates.*

Nonclaim L.11 (No proof by certificate). *Supplementary ledgers, manifests, and adversarial consistency checks support reproducibility; they do not substitute for the semantic theorem chain.*

M Bounded cross-corpus stability

The inventory in this appendix is the frozen source register of the inherited bounded-corpus analysis. Its document and page totals are not a fresh census of the enlarged corpus. The primitive-generation reference edition is separately identified in Section 23 and the bibliography.

Table 24: Cross-corpus inventory summary.

Inventory	Documents	Pages	Disposition
Bounded reference corpus	80 PDFs	5,099	Seventy-nine unique document payloads; one duplicate group is byte identical.
Selection-relevant theorem outputs	29 rows	—	Every row is classified on both benchmarks and by target relation.
Current-target selectors	0	—	No registered theorem eliminates one benchmark or identifies their carrier traces.
Explicit internal witnesses	2	—	q_W and q_{3N} survive every applicable current-target import.

Table 25: Supplementary corpus records.

Supplementary file	Function
support/CROSS_CORPUS_SOURCE_REGISTER.csv	Document-level title, page, hash, and provenance register.
data/cross_corpus_constraint_matrix.csv	Twenty-nine same-target, projection, target-fiber, future-target, and forbidden-selector dispositions.
data/residual_survivor_matrix.csv	Side-by-side carrier, operator, symmetry, low-energy, matching, pattern, and terminal status of q_W and q_{3N} .
data/corpus_selector_certificate.json	Bounded counts, benchmark survival, and no-current-selector summary.
data/terminal_roc_certificate.json	Fixed terminal role count, live-residue lower bound, lawful terminal action, and over-residue conclusion.

Within the declared bounded corpus, both benchmark carriers remain live and no current-target selector is present. Ultraviolet-sensitive evidence with independent standing therefore defines a distinct target extension rather than a same-target refinement of PCR-5.

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